Implementation and Evaluation of the 1990 Simplified Jordan Sorting Algorithm

Master's Thesis

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Current implementation and
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Abstract

Jordan sorting asks for the order of the intersection points of a simple curve and a line along the line, given their order along the curve. Hoffmann et al. (1986) and Fung et al. (1990) showed that this can be done in linear time when the underlying dynamic lists support the required finger-search operations and cost bounds. This thesis presents an executable Python reconstruction of the 1990 simplified algorithm. A single sorting procedure runs on two interchangeable backends: ordinary Python lists and a heterogeneous red–black finger tree. For the tree backend, a manual amortized analysis gives linear total work on valid inputs under unit-cost comparison and machine-word assumptions, with expected dictionary lookup and amortized growth.

In a paired experiment on 302 fixed inputs of lengths 32–513 from five input families, the tree backend, timed without the instrumentation used for operation counting, was faster on 44 inputs and the list backend on the remaining 258; the median tree/list ratio was 1.257 (range 0.590–1.518). The median ratio fell from 1.373 for lengths 32/33 to 1.037 for 512/513, and the tree's largest advantages occurred on heavily nested inputs that require almost no splitting. The results describe these input families on one machine; the experiment does not isolate the cost of individual operations.

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1. Introduction

1.1. Motivation and Scope

Jordan sorting recovers coordinate order from the traversal order of a simple curve's intersections with a line. The geometric restrictions permit linear-time algorithms [1, 2] when their specialized dynamic-list operations meet the required cost bounds. The problem arises in clipping a simple polygon against a rectangular window: intersections encountered along the polygon boundary must also be ordered along the window's edges. Fung et al. use Jordan sorting as a step in their polygon-clipping algorithm [2].

Replacing these structures with ordinary Python lists can preserve semantics while changing complexity. A structure with a better asymptotic bound may still run more slowly at practical input sizes. This thesis examines the relationship between cost bounds and observed runtime by combining algorithm reconstruction, cost analysis and measurement in an algorithm-engineering study [3, 4].

Since the September progress report, which presented the ordinary-list implementation and its runtime baseline, the scope has expanded to include a finger-tree backend, its amortized analysis and a direct comparison of the two backends.

Both backends share one Python core for the 1990 procedure. With either backend, the core maintains its own sorted output, pair forests and sibling ownership. Inputs are pre-certified valid Jordan sequences with distinct comparable values. Recognition, invalid-input behavior and geometric applications lie outside the study.

1.2. Research Questions

RQ1: Reconstruction and correctness: Can the 1990 procedure be implemented so that it computes the sorted order from its own maintained state, using either ordinary lists or a finger tree, while preserving the invariants of the pair hierarchy?

RQ2: Observed runtime: How do the two backends compare in running time on a fixed set of validated inputs under a paired measurement protocol?

RQ3: Structure and cost: Which properties of the implemented operations yield the linear total-work bound, and how do the concrete operation costs relate to the measured runtime?

The correctness question is addressed through source-linked reasoning and bounded execution checks. The runtime comparison uses paired measurement and independent archive validation. The final question brings these analyses together to relate the implemented operations to the measured runtime.

1.3. Contributions and Findings

The reconstruction specifies the choices of orientation, geometric endpoint and first point. Its tree provides stable identities, endpoint insertion, local split/join and bounded header ownership updates. A manual argument links reversed-pointer representation, locality and two potentials to linear total source-level work. This bound assumes unit-cost comparisons and machine-word operations, expected dictionary lookup and amortized growth.

The experiment compares ordinary lists with the uninstrumented tree on 302 unique inputs from five sources: common, targeted, structural, geometric and seeded. All inputs passed qualification, and the complete formal attempt produced correct outputs throughout. The tree is faster on 44 inputs. The median tree/list ratio is 1.257, with range 0.590–1.518.

Ratios decrease across the measured size bands. Within common, more heavily nested categories favor the tree more often. The heavy inputs have at most one split and no two-sided split, whereas the other sources extend coverage to interior and repeated splitting.

1.4. Reading Guide

Chapters 2 to 4 present the background, algorithm reconstruction and implementation. Chapters 5 to 7 describe the experimental method, report the measurements and discuss their limits. Chapter 8 answers the research questions. Chapter A develops the proof, and chapter B indexes its sources and supporting evidence.

2. Background

2.1. Jordan Sequences and the Sorting Problem

Let C be a simple plane curve that crosses the x-axis wherever it meets the axis. Listing the abscissas of the intersection points in the order in which they occur along C gives a sequence

$$z_1, z_2, \dots, z_n. \tag{2.1}$$

Such a realizable sequence is called a *Jordan sequence*. The Jordan sorting problem is to recover the order of the same points along the x-axis, that is, their increasing coordinate order, from their order along the curve [1, p. 170]. The 1990 presentation uses the term Jordan curve for both open and closed simple curves; the property needed here is the absence of self-intersections [2, pp. 85–86].

This formulation gives two distinct orders on one set of points:

Curve order is the input order $z_1, \dots, z_n$ obtained by traversing C.
Axis order is the output order obtained by moving from left to right along the x-axis.

The simplicity of C constrains which permutations of the coordinates are geometrically realizable. The implementation studied in this thesis specializes this model to distinct, mutually comparable coordinate values, as stated in section 4.2.

The problem arose from geometric applications. The 1986 paper relates it to ordering the intersections of a simple polygon with a line and to ordering intersections of two Jordan curves. The 1990 paper develops polygon clipping as its main application and also notes the role of Jordan sorting in simple-polygon triangulation [1, pp. 170–171, 2, pp. 85–86]. In these settings, intersection points can be obtained naturally in boundary or curve order, while the next geometric stage requires their order along another line or boundary.

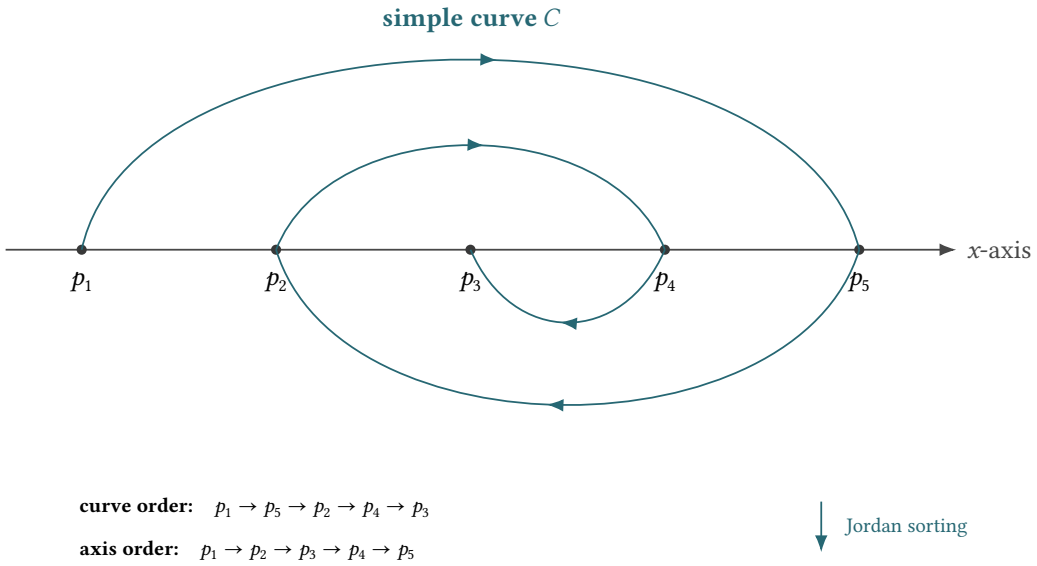


Figure 2.1.: Illustration of the Jordan sorting problem. The simple curve encounters the five intersections in curve order $(p_1, p_5, p_2, p_4, p_3)$, whereas their axis order is $(p_1, p_2, p_3, p_4, p_5)$. Jordan sorting recovers the axis order from the curve order.

2.2. The Structural Restriction Behind Jordan Sorting

Assume, after reflection about the x-axis if necessary, that the first arc lies above the axis. Consecutive intersections delimit alternating arcs above and below it. The pairs

$$\{z_1, z_2\}, \{z_3, z_4\}, \dots \tag{2.2}$$

belong to the upper side, while

$$\{z_2, z_3\}, \{z_4, z_5\}, \dots \tag{2.3}$$

belong to the lower side. Each pair determines the open interval between its two coordinates on the axis.

Two intervals from the same side cannot cross. If one endpoint of one such pair lies strictly inside another pair's interval, then its other endpoint must lie there as well; otherwise the corresponding arcs would intersect. Consequently, same-side intervals are disjoint or nested. Replacing the smaller endpoint by a left parenthesis and the larger by a right parenthesis therefore produces one properly nested parenthesis system for each side [1, pp. 172–173].

The containment relation gives an ordered forest for each parity family. A pair's parent is its smallest same-family enclosing pair, and siblings are ordered by their left endpoints. Adding a dummy pair

spanning $(-\infty, +\infty)$ turns each forest into a rooted ordered tree. These are the upper and lower family trees used by both historical algorithms [1, p. 173, 2, p. 87]. The full notation and executable pair model used in this thesis are developed in section 3.2.

This structure is what makes specialized algorithms possible for valid inputs. While processing points in curve order, the algorithms maintain both a sorted list of the processed coordinates and the two family trees. Inserting the next pair into its family identifies a local position for the new coordinate in the sorted list [1, pp. 173–175]. Efficient sorting therefore depends on maintaining and splitting these ordered family lists.

2.3. The 1986 Linear-Time Algorithm

Hoffmann et al. [1] gave an algorithm that both recognizes and sorts Jordan sequences in $O(n)$ time. It processes the sequence from left to right in curve order, maintaining a doubly linked sorted list and the upper and lower family trees. When a new interval would violate the required nesting, the recognition algorithm aborts; if all points are processed, the maintained list is the sorted result [1, pp. 173–175].

The central data-structural operation is splitting a family list around the new interval. A circular doubly linked representation permits local insertion but can require a scan proportional to the smaller side of a split. Across a sequence of splits this yields an $O(n \log n)$ implementation, no better asymptotically than general sorting [1, pp. 175–176]. The paper obtains its linear bound by replacing those lists with circular level-linked $(2, 4)$ trees.

A level-linked $(2, 4)$ tree stores list items at the leaves of a balanced multiway search tree and connects nodes at equal height. These links support finger searches whose cost depends on the shorter circular distance between the starting item and the target. Together with tree splitting, concatenation, and an amortized analysis, the representation supports the family-list operations within $O(n)$ total time [1, pp. 177–183]. Thus the 1986 result couples two ideas:

1. reduce Jordan recognition and sorting to incremental family-list manipulation; and
2. realize that manipulation with a specialized level-linked data structure.

The second component is essential to the published linear-time bound.

2.4. The 1990 Simplification

Fung et al. [2] retained the linear theoretical bound while simplifying both the primary structure and the operations required of it. A locality lemma constrains re-entry into the region of one parent and its direct children, called a local family in that paper. Its exact grouping rule stores those children in one or two contiguous sibling lists for that parent. As a result, the sorting algorithm needs only three abstract list operations:

1. create a singleton list;
2. insert an item at the front or back of a known list; and
3. split a list at a specified item or key.

In particular, the simplified formulation avoids the 1986 operation that extracts a middle section and concatenates the two remaining ends [2, pp. 86, 88–90].

The algorithm again processes intersections incrementally. It maintains a sorted list, upper and lower family trees, and sibling lists encoding the direct-child order. Neighboring processed coordinates locate the relevant same-family boundaries; insertion of the new pair and any necessary sibling split then determine where the new point enters the sorted list. The exact Step 1/2/3 reconstruction used in this thesis, including the symmetric orientation and endpoint clarifications, is the subject of chapter 3.

For the theoretical implementation, the 1990 paper proposes heterogeneous finger trees. It states constant amortized time for singleton-list creation and boundary insertion, and a logarithmic-in-the-smaller-side amortized bound for splitting. A potential argument charges the split cost across the operation sequence, giving $O(m)$ total time for m list operations and hence an $O(n)$ Jordan-sorting bound [2, pp. 90–91]. The list representation and its operation costs are essential to this bound.

The 1990 paper also states that the sorting method extends to recognition and to the error-correcting Jordan-sorting task used in the polygon-triangulation algorithm of Tarjan and Van Wyk [5], and that neither extension changes its $O(n)$ bound [2, p. 91].

2.5. Related Work and Positioning

2.5.1. Ordered Lists and Finger Search

The specialized list backends used for Jordan sorting belong to a broader line of work on ordered-list representations with fingers and locality-sensitive access. Brown and Tarjan [6] study linear lists with movable fingers, and Brown and Tarjan [7] analyze a 2–3-tree representation for sorted lists

whose costs reflect locality of reference. Huddleston and Mehlhorn [8] develop level-linked weak B-trees for arbitrary sequences of sorted-list updates. The level-linked $(2, 4)$ trees used by the 1986 Jordan-sorting algorithm draw on this line of data-structure research.

These works provide context for the historical linear-time theorem. The ordinary-list backend retains explicit scan, copy, slice and rebinding costs; the heterogeneous backend instead follows the finger-tree approach and is analyzed against its actual source representation in chapter 4.

2.5.2. Jordan Sorting and Geometric Applications

The first symposium publication of the linear-time Jordan-sorting result appeared in 1985 [9]; the 1986 journal article used throughout this thesis gives the expanded algorithm and data-structure account. The later simplification connected Jordan sorting to polygon clipping [2], while the error-correcting task appears in the polygon-triangulation algorithm of Tarjan and Van Wyk [5]. Subsequent work by Bar-Yehuda and Chazelle [10] used a generalized form of Jordan sorting together with point location and interval trees to triangulate collections of disjoint Jordan polygonal chains.

Other work pursued alternatives to the 1986 and 1990 dynamic-list algorithms. Aurenhammer [11] related Jordan sorting to convex-hull computation for certain non-simple polygons. Sojka [12] instead emphasized simpler algorithms intended to be practical at moderate input sizes, including an $O(n \log n)$ worst-case method and an $O(n \log \log n)$ worst-case method. A later average-case study analyzed a new simple algorithm under an explicit random-Jordan-sequence model, identified conditions for linear expected time, and supplemented the analysis with simulation [13]. These alternatives address different trade-offs between structural complexity, worst-case bounds and practical performance. The present study focuses on reconstructing the 1990 procedure.

The geometric applications explain why curve order is a useful input format. The present thesis studies the valid-input sorting step using the existing heterogeneous finger-tree approach.

2.6. Position and Scope of This Thesis

The implementation and evaluation distinguish three tasks:

Geometric validity. A candidate sequence is checked against the Jordan-validity model used for certification before the paper core is timed.

Valid-input sorting. The reconstructed paper control flow maintains its own partial order and recovers the final output from that state.

List representation. Sibling lists use ordinary Python lists or the implemented heterogeneous finger-tree backend, with separate cost analyses.

The empirical study uses oracle-certified valid inputs under the protocol in section 5.1. Certification, sorting and output checking are connected through the interfaces described in section 4.4.

The ordinary-list backend provides a direct representation of the sibling lists, with costs for scanning, slicing, copying and rebinding their contents. The heterogeneous backend requires a source-level account of its physical representation, locality and amortized costs. These two implementations form the comparison developed in the following chapters, with its limits discussed in sections 7.2 and 7.4.

3. Algorithm Reconstruction

3.1. From the Paper to the Implementation

The Jordan sorting problem starts with the intersections of a simple curve and the x-axis in curve order,

$$z_1, z_2, \dots, z_n, \tag{3.1}$$

and asks for the same points in increasing x-coordinate order. The 1990 algorithm of Fung et al. [2] processes this sequence incrementally while maintaining a sorted list and two ordered family trees. Its presentation is compact: the increasing case is described explicitly, the decreasing case is called symmetric, and some endpoint details are left implicit. This chapter states the executable reconstruction used in the thesis.

The reconstruction distinguishes three sources for its rules:

Explicit source rule. A definition or step is stated directly in the 1990 paper, such as the upper/lower family trees, predecessor and successor boundary searches, and increasing Step 3.

Symmetric reconstruction. A decreasing rule is obtained by reflecting the explicit increasing rule: predecessor and successor roles, left and right split sides, child extremes, and insertion direction are exchanged.

Executable clarification. A detail is fixed so that the compact paper description becomes unambiguous code. The geometric child endpoint and the odd-index z_1 output-anchor rule fall into this category.

The executable clarifications are implementation-level interpretations supported by the related 1986 algorithm of Hoffmann et al. [1], reflected counterexamples and bounded exhaustive validation over oracle-certified inputs.

The chapter also separates algorithmic control flow from data-structure complexity. The 1990 paper obtains its linear bound with heterogeneous finger trees. Both thesis backends realize the same high-level operations. Ordinary Python lists incur the quadratic copying lower bound in section 4.6; the finger-tree backend has a separate source-linked representation and amortized-cost argument there.

3.2. Pairs, Intervals, and the Two Families

Input values are assumed to be distinct and mutually comparable. For each $i \geq 2$, define the curve-order pair

$$P_i = \{z_{i-1}, z_i\}. \tag{3.2}$$

The two stored endpoints retain curve order. Their geometric interval is described separately by

$$\ell_i = \min\{z_{i-1}, z_i\}, \qquad r_i = \max\{z_{i-1}, z_i\}. \tag{3.3}$$

A pair P_i encloses a finite point x when

$$\ell_i < x < r_i, \tag{3.4}$$

and it encloses another pair P_j when

$$\ell_i < \ell_j < r_j < r_i. \tag{3.5}$$

After reflecting the curve about the x-axis if necessary, the pairs with even end index belong to the upper family and those with odd end index belong to the lower family:

$$P_2, P_4, P_6, \ldots \in \mathscr{F}^+, \qquad P_3, P_5, P_7, \ldots \in \mathscr{F}^-. \tag{3.6}$$

For pairs in one family, simplicity of the curve implies that their intervals are disjoint or nested. The enclosure relation therefore forms a forest. Each set of siblings is ordered from left to right by increasing ℓ_i.

The reconstruction adds a distinct dummy pair to each family. Each dummy conceptually spans $(-\infty, +\infty)$ and becomes the parent of every pair that has no finite enclosing parent. The two forests are thereby represented as rooted ordered trees, but the upper and lower dummy identities remain separate.

3.2.1. Families and Sibling Lists

Here *parity family* means the entire upper or lower forest $\mathscr{F}^+$ or $\mathscr{F}^-$. For a particular owner p, its *owner region* lies between its arc and its direct children's arcs; a *sibling group* is one stored list of those direct children. The local "family" in the locality argument of Fung et al. [2, pp. 87–88] consists of a parent and its direct children, not the whole parity forest.

The exact grouping invariant G is as follows. Let P_j have the largest end index among the finite members of $\{p\} \cup \text{children}(p)$. If $P_j = p$, all children form one group. Otherwise, the nonempty groups are the children with $r_h \leq z_j$ and those with $\ell_h \geq z_j$, respectively. Empty groups are omitted;

an owner with no children has no group. The dummy is excluded when selecting j. Thus "at most two lists" is necessary but insufficient: the cut must be the endpoint of this particular latest child. The locality and preservation proof is given in section A.2. This locality is what reduces the update to three abstract list operations:

1. create a singleton sibling list;
2. insert a pair at a known list boundary; and
3. split a sibling list at the value of the new point.

The algorithm never requires arbitrary insertion into the middle of a sibling list. A split may produce an empty side, but an empty side is not retained as a live sibling list.

3.3. Maintained State and Prefix Invariant

After processing $z_1, \dots, z_k$, the conceptual state contains:

- a partial-order list S_k containing exactly those k points in increasing value order;
- every finite pair $P_2, \dots, P_k$ in its parity family;
- one parent and one sibling-list owner for every finite pair; and
- the two family-specific dummy roots.

The central prefix invariant is

$$S_k = \mathrm{sort}(z_1, \dots, z_k). \tag{3.7}$$

Here `sort` is mathematical notation for the required order, not an operation invoked by the paper core. The maintained list is bounded by identity-based negative- and positive-infinity sentinels so that predecessor and successor queries are defined at both ends.

The exact grouping invariant G, the family-tree invariant and equation (3.7) are updated together. Pair insertion locates the interval in which z_i belongs, sibling splitting transfers any newly enclosed children, and the resulting family boundary determines the local insertion anchor in S_{i-1}.

3.4. Initialization

Lengths zero, one, and two are handled directly by fixed-size comparison logic. For $n \geq 3$, the incremental state is initialized by ordering z_1, z_2, z_3 with fixed-size comparison logic and creating

$$P_2 = \{z_1, z_2\} \qquad \text{in the upper family,} \tag{3.8}$$

$$P_3 = \{z_2, z_3\} \qquad \text{in the lower family.} \tag{3.9}$$

The upper dummy owns a singleton sibling list containing P_2, and the lower dummy owns one containing P_3. Thus S_3 satisfies equation (3.7), both parity families have a root, and every finite pair has exactly one owner before the main loop begins.

3.5. Boundary Selection

Iteration $i \geq 4$ creates P_i in the family determined by the parity of i. Steps 1 and 2 inspect the immediate neighbors of z_{i-1} in S_{i-1}.

3.5.1. Step 1: Predecessor-Side Boundary

Let v be the predecessor of z_{i-1}. If i is odd and $v = z_1$, replace v by the predecessor of z_1. If the resulting neighbor is the negative sentinel, let A_i be the dummy pair of the current family. Otherwise, let A_i be the already processed same-family pair containing v.

3.5.2. Step 2: Successor-Side Boundary

Let w be the successor of z_{i-1}. If i is odd and $w = z_1$, replace w by the successor of z_1. If the resulting neighbor is the positive sentinel, let B_i be the current family dummy. Otherwise, let B_i be the processed same-family pair containing w.

A processed point can belong to both an upper and a lower pair. The parity of i selects the relevant incident pair without using a global rank map. The special treatment of z_1 is necessary because z_1 belongs to P_2 but to no lower-family pair. These two neighbor adjustments select family-tree boundaries; they are distinct from the output-anchor adjustment introduced in section 3.8.2.

3.6. Step 3: Increasing Orientation

Assume first that $z_{i-1} < z_i$. Let $N = P_i$ denote the new pair. The predecessor-side pair A_i controls where N enters the family tree, while the successor-side pair B_i controls whether an existing sibling list must be split.

3.6.1. Step 3(a): Insert the New Pair

If A_i encloses z_{i-1}, then N becomes its child in a new singleton sibling list. Otherwise, A_i is the last item of its sibling list; N is inserted immediately after A_i and inherits the same parent. In either case, N obtains exactly one parent and one sibling-list owner.

3.6.2. Step 3(b): Split Off Newly Enclosed Children

If B_i encloses z_{i-1}, N acquires no child list. Otherwise, B_i is the first item of an existing sibling list L. Splitting L at z_i produces a left side and a right side. For a valid input, no pair may straddle the split value: both endpoints of each child lie on one side. The left side, whose pairs are newly enclosed by N, becomes a child list of N; the right side remains with the previous owner of L.

3.6.3. Step 3(c): Insert the New Point

If N acquired no children, the base output anchor is z_{i-1}. Otherwise, let C be the rightmost child pair of N and choose the geometric right endpoint $r(C)$ as the base anchor. After the possible odd-index adjustment in section 3.8.2, insert z_i immediately after the output anchor.

3.7. Step 3: Decreasing Orientation

For $z_i < z_{i-1}$, the reconstruction uses the order reflection of the increasing case. This is the executable meaning assigned to the 1990 paper's statement that the other case is symmetric.

Step 3(a) now uses B_i. If B_i encloses z_{i-1}, N becomes its child in a singleton sibling list. Otherwise, B_i is the first item of its list; N is inserted immediately before it and inherits its parent.

Step 3(b) uses A_i. If A_i encloses z_{i-1}, no split is needed. Otherwise, A_i is the last item of its sibling list. The list is split at z_i; the right side is transferred to N, while the left side remains with the old owner.

For Step 3(c), the no-child base anchor is again z_{i-1}. With children, let C be the leftmost child pair of N and use its geometric left endpoint $\ell(C)$. After the reflected odd-index adjustment, insert z_i immediately before the output anchor.

The complete reflection is summarized in table 3.1.

Table 3.1.: Increasing and decreasing Step 3 roles in the executable reconstruction.

Operation	Increasing	Decreasing
Step 3(a) boundary	predecessor-side A_i	successor-side B_i
Boundary insertion	after last sibling	before first sibling
Step 3(b) boundary	successor-side B_i	predecessor-side A_i
Acquired split side	left	right
Child extreme	rightmost child	leftmost child
Geometric anchor	right endpoint	left endpoint
Output insertion	after anchor	before anchor

3.8. Executable Anchor Clarifications

The point insertion in Step 3(c) is where the compact paper notation requires the most care. Two independent distinctions are necessary.

3.8.1. Curve Endpoint Versus Geometric Endpoint

The 1990 increasing step refers to the rightmost child $\{z_{m-1}, z_m\}$ and inserts the new point after z_m. A pair record, however, stores its endpoints in curve order, and its second curve endpoint is not necessarily its geometric right endpoint. Reading z_m mechanically as the second stored endpoint fails on valid reflected orientations.

For example, consider

$$(3, 2, 1, 4). \tag{3.10}$$

At $i = 4$, the acquired child is $P_2 = \{3, 2\}$. Inserting 4 after the second curve endpoint 2 would yield $(1, 2, 4, 3)$. Inserting after the child's geometric right endpoint 3 yields the correct order. The decreasing counterpart

$$(2, 3, 4, 1) \tag{3.11}$$

similarly requires insertion before the child's geometric left endpoint 2, not before its second curve endpoint 3.

The reconstruction therefore uses $r(C)$ for the increasing branch and $\ell(C)$ for the decreasing branch. This choice agrees with the earlier family-list account of Hoffmann et al. [1], in which both endpoints occur in sorted family lists and the new point is inserted next to the located boundary item. This is the geometric-endpoint clarification used in the implementation.

3.8.2. Odd-Index z_1 Output Anchor

Steps 1 and 2 compensate for the fact that z_1 has no lower-family pair. That boundary-pair correction does not cover every final output insertion. The 1986 description records a separate anomaly: on an odd iteration, if z_1 lies between the ordinary family-list boundary and the new point, the new point is inserted next to z_1 [1, p. 175].

The executable increasing rule is

$$\text{if } i \text{ is odd and } a_i < z_1 < z_i, \quad a_i \leftarrow z_1, \tag{3.12}$$

where a_i is the geometric base anchor. The decreasing reflection is

$$\text{if } i \text{ is odd and } z_i < z_1 < a_i, \quad a_i \leftarrow z_1. \tag{3.13}$$

For the decreasing sequence prefix

$$(1, 2, 3, 4, 6, 7, 0), \tag{3.14}$$

the ordinary geometric base anchor at $i = 7$ is 2. Inserting 0 before 2 would leave 1 on the wrong side. Because $0 < z_1 = 1 < 2$, the output anchor is changed to z_1, and insertion before it gives the correct prefix order. The reflected sequence $(6, 5, 4, 3, 1, 0, 7)$ exercises equation (3.12). These cases demonstrate why boundary selection and output-anchor correction must be modeled as separate stages.

3.9. Ownership Transfer and Failure Semantics

A sibling split changes several linked facts at once: the old list ceases to exist, up to two nonempty output lists are created, the retained side stays with the old parent, and the acquired side moves under N. Each moved pair must belong logically to exactly one output sibling group and have that group's owner as its parent. The ordinary backend updates per-pair ownership fields, whereas the tree backend updates headers without rebinding every leaf. Conservation and unique ownership are required after a successful transition.

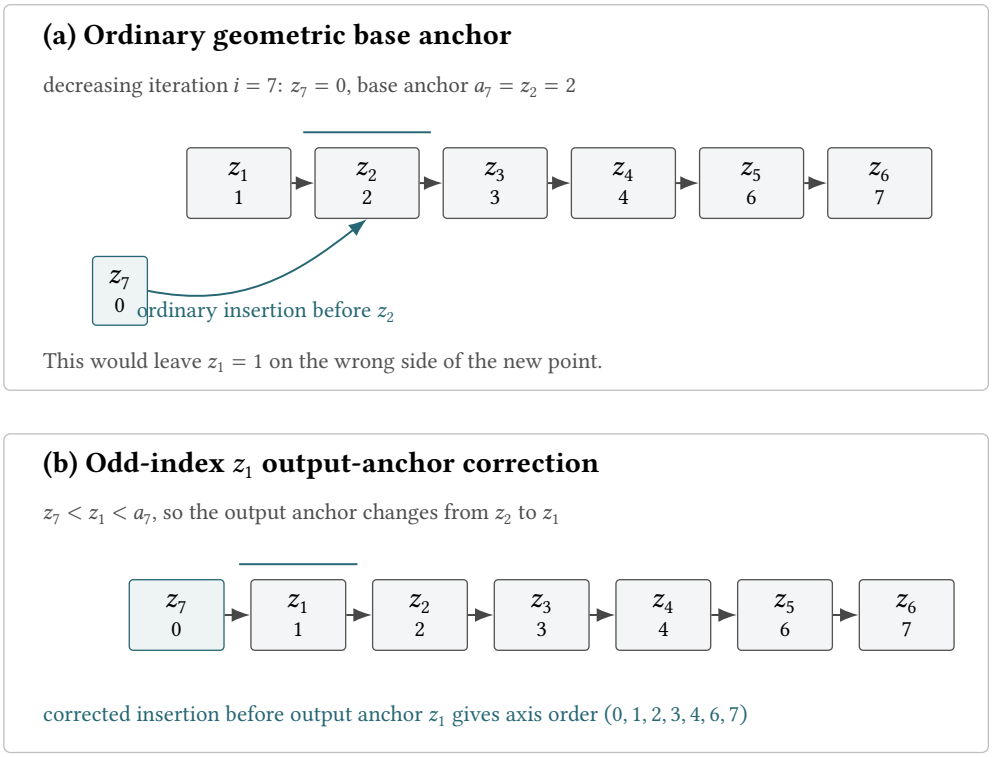


Figure 3.1.: Local output-anchor selection in decreasing Step 3(c) for the prefix $(1, 2, 3, 4, 6, 7, 0)$. The ordinary geometric base anchor is z_2. Because the odd-index condition $z_7 < z_1 < z_2$ holds, the output anchor is corrected to z_1 before z_7 is inserted.

The ordinary backend first forms a split plan without changing live state; its commit checks the plan, parent and partition before updating ownership. The tree instead uses a planned transfer source, path-local splitting and header adoption, without materializing that list-based plan. Both must restore the logical ownership conditions before completing the iteration. In the revised implementation, an update or postcondition error terminates sorting. Updates already performed are not rolled back, and the failed state must be discarded. No partially updated state is accepted as a completed iteration or returned as a sorting result.

The earlier implementation used rollback on these error paths. The version history is described in section 4.1, and the concrete update operations in section 4.3.3.

3.10. End-to-End Procedure

The full valid-input sorting procedure is:

1. Materialize the input once and handle lengths zero, one, and two directly.
2. For $n \geq 3$, initialize S_3 by fixed-size ordering of the first three points, then create P_2, P_3, the two dummies, and their singleton sibling lists.
3. For each $i = 4, \ldots, n$:
   Step 1. Compute predecessor-side boundary A_i.
   Step 2. Compute successor-side boundary B_i.
   Orient. Choose increasing or decreasing Step 3 from the comparison of z_{i-1} and z_i.
   Update. Insert P_i, split and transfer any enclosed children, and insert z_i using the resulting geometric output anchor.
   Commit. Advance the processed prefix only after the output insertion succeeds.
4. Return the values obtained by traversing S_n.

At the end of each successful iteration, the required online conditions are:

- S_i contains each processed point exactly once and is strictly increasing;
- every processed finite pair exists in the family dictated by its parity;
- every finite pair is a member of exactly one live sibling group, whose owner is its parent;
- sibling order follows geometric left endpoints, parent chains are acyclic, and each chain reaches the appropriate dummy root; and
- split outputs are disjoint and preserve the input pair set.

The final return value is recovered from the maintained order S_n. The certification wrapper that establishes the input precondition is described in section 4.4.

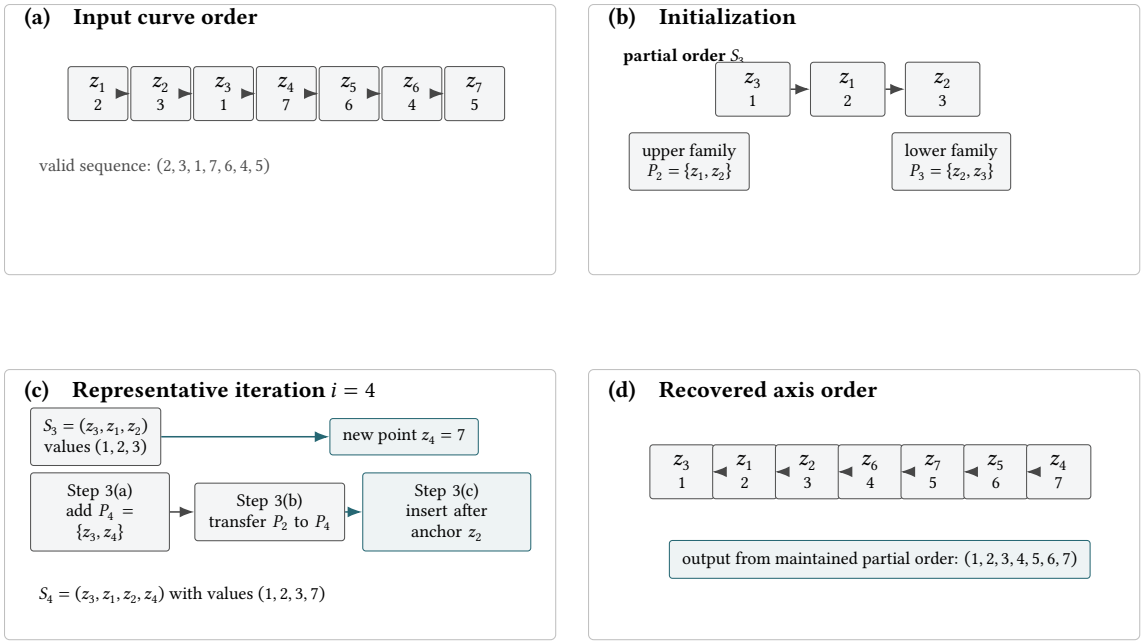


Figure 3.2.: Running example on the valid sequence $(2, 3, 1, 7, 6, 4, 5)$. The panels show the input, initialization, the representative update at $i = 4$, and the recovered axis order. At $i = 4$, P_4 acquires P_2 in Step 3(b), then Step 3(c) inserts z_4 after the geometric anchor z_2.

3.11. Correctness and Operation Costs

The source-level correctness argument rests on the nesting property of same-family pairs and the locality lemma: the boundary pairs found beside z_{i-1} identify where P_i belongs, and newly enclosed children form a contiguous split side. The executable reconstruction turns those facts into the prefix and ownership invariants above. Focused counterexamples fix the ambiguous anchors; mirrored tests check the decreasing reconstruction; and full state diagnostics compare the maintained order and family forest with a deterministic replay.

The cost of these steps depends on the backend. Hoffmann et al. [1] use level-linked search trees, while Fung et al. [2] reduce the required operations to singleton-list creation, boundary insertion, and splitting, then implement them with heterogeneous finger trees. The ordinary backend preserves those abstract operations using Python lists. A sibling split scans and materializes list contents, both insertion boundaries copy the current list, and ownership rebinding can depend on list size. Membership and ancestor checks can also scan lists during sorting. The tree backend instead uses the physical operations described in chapter 4, where the costs of both representations are analyzed.

4. Implementation

4.1. Architecture and Versions

The code separates an oracle-backed reference pipeline from the shared-core reconstruction of Fung et al. [2]. The reference function `simplified_jordan_sort` certifies the input, constructs family trees, statistics and a trace, and returns the oracle's sorted output. In contrast, `paper_jordan_sort_valid` executes the reconstructed algorithm and returns `state.partial_order.to_list()`. Its core does not use the oracle's output or a global rank map. It assumes a pre-certified valid Jordan sequence with distinct, mutually comparable values.

Both current backends propagate errors without rollback. The paired timings in chapter 6 measure these programs; the earlier rollback-enabled study is documented in its historical section. Source bindings are listed in chapter B.

Chapter 3 specifies initialization, both orientations of Step 1/2/3, and executable anchor clarifications. Here the focus is state representation and maintenance. Both backends share the sorting loop, while their representations determine how they perform each list operation.

4.2. Data Representation

An immutable `PointRef(paper_index, value)` separates one-based paper identity from a point's comparable value. A `PairRecord` retains curve endpoint IDs and family membership; geometric endpoints are obtained by comparing their values. The distinction and its consequences for output anchors are specified in section 3.8.1.

`PaperJordanState` holds these records, the processed-prefix length, the partial order, the end-index-to-pair mapping, the sibling backend and two family roots. It also holds a fixed execution policy, optional trace and counters, and mandatory stage results. Sorting and diagnostics use the same runner and mirrored orientation branches.

4.2.1. Partial Sorted Order

`SortedOrderList` is doubly linked, with identity-based infinity sentinels and a point-ID-to-node mapping. Neighbor access and insertion use local links. Before mutation, insertion checks that the value lies strictly between its adjacent values. Final traversal excludes the sentinels and returns the original values. Mapping access uses Python dictionaries and is charged at its expected cost.

4.2.2. Sibling Lists and Ownership

Each `SiblingList` stores ordered finite pair IDs and one owner. A pair caches its parent and containing-list IDs, together with its ordered child-list IDs. Every owned finite pair occurs in exactly one live list, whose owner equals its parent. A parent has at most two ordered child lists; finite parent chains must reach the same-family dummy without cycles. The family dummies are roots, not finite list items.

The ordinary backend retains this eager representation. The shared core uses boundary-list, owner and extreme-child access interfaces instead of reading those fields directly. General pair-liveness checks do not require the pair to be a list endpoint.

The tree backend stores stable pair handles in external leaves, with internal separator nodes, two endpoint fingers and a list header. Outer rib pointers point toward the root; interior pointers point toward children. Separators store the maximum left key. Rank is black height excluding external black leaves. For internal nodes, span counts all leaves in the INTERIOR role, the inward branch on a rib, and zero in the ROOT role. External leaves always have span one, including a singleton root. The header separately stores total size. Ownership belongs to the header and at most two parent child slots, avoiding per-leaf rebinding. Boundary slots only identify designated endpoints; they are not arbitrary-owner lookup caches.

4.3. Core–Backend Interface and Update Semantics

4.3.1. Initialization, Boundaries and Stage State

The initialization in section 3.4 materializes all point records but orders only the first three values; shorter inputs use fixed-size comparisons. Subsequent iterations use the predecessor and successor boundaries defined in section 3.5. The backend handles singleton creation, boundary insertion, splitting and ownership transfer; the partial-order structure handles point insertion.

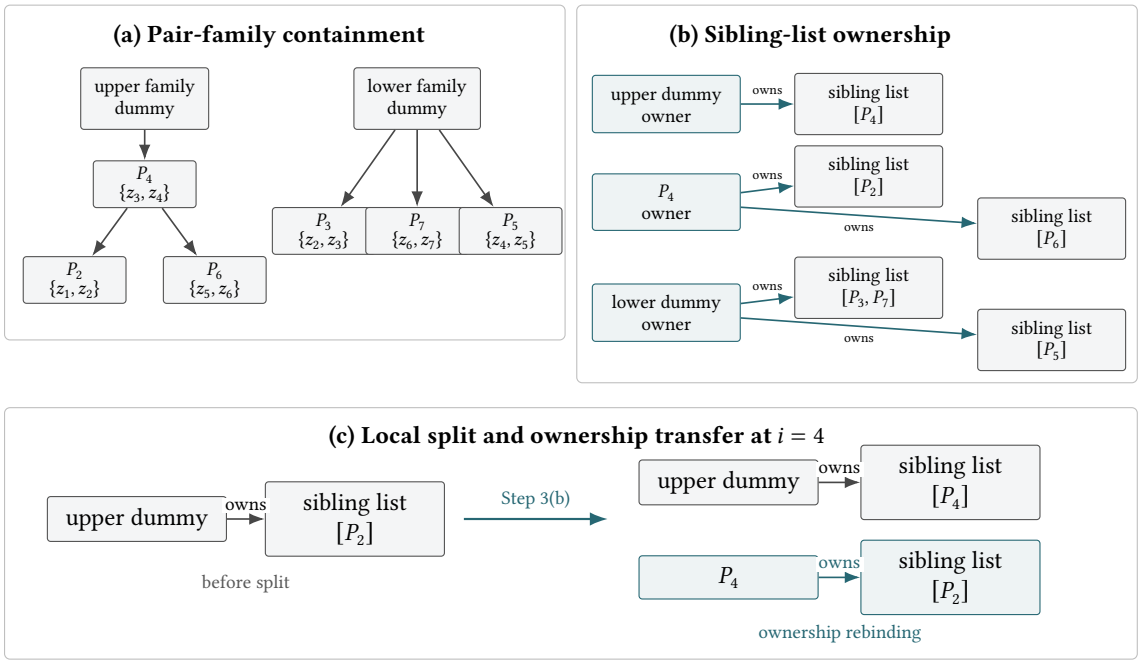


Figure 4.1.: Pair families and sibling-list ownership in the example of figure 3.2. Step 3(b) transfers P_2 from a list owned by the upper dummy to a child list of P_4.

Boundary selections and Step 3 results are stored even with trace disabled. Later stages consult them to reject incorrect order, repeated stages and inconsistent parameters. They are execution state, not merely optional logging. Optional backend-work observations are stored separately from these mandatory results; complete logical snapshots belong to the audit path.

4.3.2. Pair Insertion and Failure Propagation

For the insertion rules in sections 3.6.1 and 3.7, the core checks the boundary, registers the new pair and calls singleton creation or boundary insertion. A later error may leave that registration in place. The exception propagates; the state is discarded rather than restored or reused. This contract applies to both orientations.

4.3.3. Split Plans and Ownership Transfer

The ordinary backend implements the partition specified in sections 3.6.2 and 3.7 using an immutable `SplitPlan`. It records the original sequence and both ordered partitions before mutation. The plan sup-

ports stale-plan rejection, order checks and pair-set conservation; these functions remain necessary without rollback.

Commit checks the plan and ownership preconditions, prepares nonempty output lists and affected child slots, retires the input list, publishes the new lists, and rebinds the pairs. Touched-state postconditions are checked before success is returned. Rejection covers stale plans, straddling pairs, ownership mismatches, cycles, unowned parents and a third child list. Errors during mutation or postcondition checks propagate without restoring earlier state.

The tree starts boundary insertion at a finger and repairs locally. A split with both sides nonempty races from both ends, detaches a component, splits its downward representation and joins by black height. Path sizes and header totals supply output sizes without enumerating leaves. Every successful split, including a one-sided result, retires its source ID and allocates fresh IDs for nonempty outputs; boundary insertion preserves its live ID. Pair handles, leaves and existing descendant groups survive transfer. Before Step 3(a), the core identifies the other group's transfer source without splitting it. Step 3(b) consumes that exact source; pending split/transfer state must clear before Step 3(c) inserts the output point.

4.3.4. Point Insertion and Completion

The backend exposes the relevant extreme child; the core selects its geometric endpoint and applies the separate odd-index z_1 anchor rule of section 3.8.2. The clarifications and counterexamples are given in section 3.8. The partial order performs the resulting adjacent insertion. Only after it succeeds does Step 3(c) advance `processed_count`.

4.4. Certification, Diagnostics and Replay

The wrapper `certified_paper_jordan_sort` runs the oracle and rejects invalid candidates before entering the pure sorter. The *minimal core* is the sorting call with optional full-state checks, operation counters and tracing disabled; necessary local checks remain active. Certification is outside this call's timing scope. An internal exception is propagated by both sorting and diagnostic entry points: neither retries, returns a partial output, nor falls back to the oracle or reference pipeline. Output recovery is reached only after the runner succeeds.

`paper_jordan_diagnostics_valid` selects `checked` and audits initialization and every completed prefix. It checks point identity and values, partial-order links and order, exact pair membership, ownership and parent chains, stage records, trace and metrics. *Same-backend replay* reconstructs a fresh state

using the same core, backend type and immutable configuration. It compares backend identity, partial order, registries, stage results, trace, work observations, metrics, and logical and physical snapshots. Unavailable backends fail explicitly rather than reverting to ordinary lists. Replay checks same-core state consistency; external sorting checks the final output.

Qualification, the pre-timing validation described in section 5.2, advances two independently initialized checked instances through the same pausable core, audits and compares every completed prefix, then performs one final fresh checked replay. A separate fresh minimal call checks the measured configuration. Snapshots reuse one audit within a validation call, never a cross-call "already checked" cache. Old stage and trace records remain checked.

Failure-injection tests separately check propagation from pair insertion and split postconditions, including checks retained in `minimal`. Public-call tests require one split attempt and no output recovery after failure.

4.4.1. Correctness Coverage

Table 4.1 indexes earlier operation, sorting and qualification checks at their recorded source snapshots. The current five-source qualification is described in section 5.2.

Table 4.1.: Selected earlier correctness evidence. Source identities are listed in chapter B.

Level	Covered behaviour and limit	Evidence record
Tree operations	53 test methods: insertion, split/adopt, both outputs updated and split again, existing descendants and stable identities. Near-nephew deletion repair has operation-level coverage; complete-sort coverage is not established.	Operation tests
Complete sorting	18 fixed balanced, skew and repeated constructions, $n = 128/129, 256/257, 512/513$; checked/minimal and three backend configurations. Correctness and resource evidence only.	Structural campaign
Common group	140 inputs qualified, including fresh minimal calls; 3,788 splits per backend, all with an empty side. This group's timing does not exercise interior splitting.	Common-set qualification
Targeted group	12 inputs qualified separately; 12 two-sided splits per backend, source size 61, smaller side 3/7/19. Later direct insertions reach only the retained side.	Split-set qualification
Seeded supplement	40 fixed inputs up to $n = 1500$: list/tree minimal and tree trace checked against sorted output; eight small cases also checked with fresh-prefix replay. 678 two-sided splits; no timing evidence.	Seeded supplement; October 20 protocol

The 18-input campaign records 460 splits, 230 two-sided splits, 666 later insertions into earlier split outputs and 212 re-splits, per checked configuration's original runs. Within it, `repeated-512`

contributes respectively 126, 63, 189 and 62 operations; its two-sided splits always have smaller side one. The 512-point balanced and skew cases split 253 siblings into 126 + 127 and 1 + 252; the 513-point cases mirror these constructions. The same eighteen inputs were subsequently qualified again and formally timed, first in the separate November extension and then in the unified study (sections 5.2 and 6.3.1).

The seeded generator extends from z_{i-1} into an axis gap such that no same-parity arc has exactly one endpoint between z_{i-1} and the new point; final axis-order ranks become the input values. With probability b it chooses uniformly among legal gaps; otherwise it follows a direction (reversed with probability 0.08), choosing its nearest or farthest legal gap with equal probability, or all legal gaps if that side is empty. Seeds 11–14 each generate sizes 128/512/513/1500 in that order, at $b = 0.05, 0.25$ for each size; seeds 21–22 each generate 32/33/64/70 at $b = 0.1$. Each seed uses one sequential Python `random.Random` stream; all forty inputs are saved and independently certified by the oracle. Thirty-eight contain a two-sided split, with maximum smaller side 13. The eight smaller inputs also receive checked validation. This seeded supplement extends the tested operation histories. The unified timing study includes its inputs of length at most 513 alongside common, targeted, structural and geometric sources. Archive locators and the source-hash distinction are in section B.1.

4.5. Execution Policies and Observation Costs

Five immutable policies control trace recording, selected operation counters and complete backend validation after split commits. `checked` enables all three; `minimal` disables them. Intermediate policies isolate observation costs. All use the same Step 1/2/3 sequence and retain stage results, necessary local checks, input materialization and output recovery. The revised implementation has no rollback under any policy.

In the instrumented tree, field interception and wrappers record work when a separate ledger is enabled. Minimal mode alone does not disable it. Setting `detailed_work=False` suppresses accumulation and reports but retains these observers. The earlier instrumented experiments used this configuration.

The current timings use the uninstrumented tree, generated from the same source with observation statements and wrappers removed and ordinary field access. Unsupported observation calls or residual ledger references cause generation to fail. Preparation precedes timing; binding methods to each new instance is timed. Local checks remain. Section 5.2 describes the equivalence and qualification evidence.

The earliest archived study timed the rollback-enabled implementation, including recovery-state preparation in `minimal`; a successful call need not actually roll back to incur that preparation. The historical scope is specified in section 5.7.

For each successful split, the ordinary backend reports an increment equal to the input-list length for both `split_items_scanned` and `split_items_copied`. These logical split-volume measures omit boundary-insertion copies, membership scans and repeated validation/materialization passes. The shared core accumulates each backend's reported work under an identified metric schema. This keeps logical volumes separate from observed field accesses and elapsed time.

4.6. Operation Costs and Complexity

In the ordinary backend, either boundary insertion constructs a new Python list containing the existing IDs and the new pair. Rear insertion is not an in-place `list.append()`. Both orientations copy work proportional to the current sibling-list length. Split additionally scans and partitions the list, materializes its outputs and rebinds pair ownership.

"Local" describes the objects a safety check concerns, not constant cost. The ordinary backend's liveness check uses `pair_ids.count(pair_id)` for membership, and backend parent checks scan the containing list at each visited ancestor. For an ancestor chain A, this contributes $O(\sum_{a \in A} |L(a)|)$, where $L(a)$ contains a, rather than a constant per ancestor. These costs remain in `minimal`; complete checked validation and prefix replay add separate untimed work.

The valid increasing input $(1, 2, \ldots, 2m)$, $m \geq 2$, exposes a quadratic copying lower bound even without split. Starting from one pair per family, boundary insertions grow the sibling lists to m and $m - 1$ while Step 3(b) skips splitting. The number of existing pair IDs copied is

$$\sum_{j=1}^{m-1} j + \sum_{j=1}^{m-2} j = (m-1)^2. \tag{4.1}$$

Thus this input family incurs $\Omega(n^2)$ copying, including in `minimal`. This lower bound accounts for insertion copying alone.

For the tree backend, the source-level proof in chapter A accounts for a different set of operations: decoding and local repair preserve the representation, exact grouping supplies legal endpoint calls, and one potential pays propagation across mixed insertion/split histories. A second, list-size potential pays the smaller-side split terms over the sorting run. Theorem A.10 includes the shared core, ownership and output work under its stated computation model.

5. Experimental Method

5.1. Study Design

The main experiment compares ordinary lists with an uninstrumented finger tree on 302 unique inputs from five sources. The input registry, exclusions, reporting rules and preparation stages were fixed before formal sampling. Both implementations use the same public Jordan sorting core. Equivalence checks, qualification and separate batch calibration precede measurement. The results use one complete session; the earlier failed attempt and the decision to restart are disclosed below.

5.2. Inputs and Qualification

The unified registry assigns every input to a source and deduplicates its rank permutation in the fixed source order. It contains no duplicates. 8 previously saved seeded inputs of length 1500 were excluded before this experiment under the resource-scope rule. No input was added or removed in response to the formal ratios.

Common contains 140 inputs at lengths 32/33, 64/65, 128/129, 256/257 and 512/513. At each length, fixed quotas select increasing strict-flat sequences, low- and medium-nesting block constructions, and sequences with a heavily nested suffix. Original and mirrored sequences remain related constructions.

Targeted contains 12 inputs at $n = 128, 129$, using `values_for` with smaller-side parameters $k = 3, 7, 19$ in both directions. The parameters were chosen to exercise interior splits, not from timing results.

Structural contains 18 saved inputs: balanced, skew and repeated shapes at 128, 129, 256, 257, 512, 513. Balanced and skew share the targeted construction; repeated uses a separate template that splits previous outputs again. Even lengths use the original direction and odd lengths the mirror, coupling parity with direction.

Geometric contains 100 inputs obtained by widening random grid trees into simple orthogonal polygons and recording the boundary order of intersections with a horizontal line. The grid height

is fixed, and each polygon uses one sampled traversal start and direction. Only the even lengths 32, 64, 128, 256, 512 occur.

Seeded contains 32 inputs generated by successive insertions into gaps allowed by the noncrossing condition, with fixed seeds, gap preferences and direction changes. Its measured lengths are 32, 33, 64, 70, 128, 512, 513. Shared random-number streams and related settings do not constitute independent repetitions.

Before timing, the instrumented and uninstrumented trees were compared on all registered inputs. Qualification then checked both the ordinary list and the uninstrumented tree. Two checked instances advanced through the same prefixes; a fresh checked replay and a fresh minimal call supplied additional checks. Expected sorted outputs were computed outside the core. This is finite execution evidence, not a proof of equivalence on all valid inputs.

5.3. Compared Configurations and Timing Scope

Configuration choice. An earlier formal experiment used the instrumented tree with detailed counting disabled. On its 140 common inputs, the tree was slower throughout, with median tree/list ratio 2.898. A separate 15-input diagnostic reduced tree medians by 37.7–52.1% when field-access instrumentation was removed. This observation motivated a configuration that excludes observation machinery from the final timing comparison. The earlier instrumented campaigns and diagnostic retain their original archives, indexed in section B.1.

The uninstrumented tree is generated from the same implementation source by removing observation statements and wrappers and replacing tracked field access with ordinary field access. Algorithmic checks and data-structure operations remain. Preparation and compilation occur once before timing. Each new tree instance binds the generated methods during its constructor, inside the timed sorting call. The instrumented tree remains available for operation counts and validation, outside the formal timing samples.

A subsequent constructor-only microbenchmark on the same machine used fixed parameters: five warm-up pairs and fifty measured pairs, with 256 retained instances per batch and alternating configuration order. Its median times were 0.420 μs per instance for the instrumented tree and 5.304 μs for the uninstrumented tree. The median paired difference was 4.881 μs. For scale, this is 0.81% of the shortest tree median across the formal inputs (605.462 μs). The timed constructor paths include automatic GC and loop/storage overhead; objects were released after timing. Nothing was subtracted from the formal sorting observations.

5.4. Timing, Calibration and Scheduling

Timing uses `time.perf_counter_ns()`. Each call includes fresh initialization, the main loop and output recovery; the batch loop and output append are also inside the interval. Expected-output comparison, explicit pre-batch garbage collection, compression and persistence are outside it. Automatic GC remains enabled.

Separate calibration tries common batch sizes $B = 1, 2, 4, 8, 16, 32, 64, 128$ in order, using three batches per backend at each attempted value. It selects the first B for which both calibration medians reach 10 ms, without consulting the ratio. Each input's B is then held fixed. Qualification allows 180 seconds per input and 1800 per source; calibration allows 600 seconds per source. Formal limits are 120 seconds per input and 1800 per source. Preparation completed for every source before sampling.

5.4.1. Scheduling and Environment

One continuous session executes common, targeted, structural, geometric and seeded in that order. Within sources, the protocol shuffles inputs with seed `20261015`. Each backend receives 5 warm-ups and 20 measured batches per input. First position alternates by input index and round, giving each backend ten first positions. No tests, compilation or other experiments run between sources. Outlier deletion and speed-based replacement are prohibited.

The environment is Apple M4/arm64 with ten logical CPUs, 16 GiB RAM, macOS 27.2 and CPython 3.12.14, with optimization disabled. GC thresholds are 700/10/10. Calibration and formal sampling require AC power and low-power mode off, checked before and after each input. Sleep is inhibited during the session. These are discrete power observations, not continuous monitoring. A preceding ten-input cross-host stability diagnostic informed the choice of host; it was separate from this experiment and did not change its protocol.

Alternating order balances first position within an input, but cannot remove all thermal or operating-system effects [14]. Repetition alone does not justify statistical inference [15].

5.4.2. First Attempt and Restart

The first attempt stopped when a post-input power check recorded battery power during the geometric source. Common, targeted and structural had completed; subsequent work stopped and all partial records were retained. The cause and exact time of the supply change were not determined.

After the failure diagnosis, the author authorized a restart from the beginning before any performance ratios were calculated. This was an explicit departure from the protocol's no-retry rule. The second attempt used the same inputs, batch sizes, source code and scheduling rules and completed all five sources. The two attempts are not joined or pooled. Archive validation confirms the recorded technical conditions; it does not erase the procedural deviation.

A later, post hoc descriptive check compares the ratios of the 170 inputs in the first attempt's three complete sources. All 170 retain the same faster backend. The absolute relative change $100|r_i^{(2)}/r_i^{(1)}-1|$ has median 0.426% and maximum 1.984%. This check is outside the protocol and excluded from the main statistics.

5.5. Aggregation and Structural Descriptions

A batch contributes one duration divided by B. For each input/backend, the 20 values yield a median, R7-interpolated quartiles and IQR. The input-level comparison is

$$r_i = \frac{\mathrm{median}_b(t_{i,\mathrm{tree},b}/B_i)}{\mathrm{median}_b(t_{i,\mathrm{list},b}/B_i)}.$$

A ratio below one favors the tree. Comparisons use unrounded values. Each unique input contributes one ratio; neither the batches nor the calls within them increase the number of independent inputs. Source, nesting-category and structural-shape summaries remain descriptive, without significance tests or confidence intervals over dependent constructions.

For each input, S_i is the number of actually performed splits in its primary checked qualification trace, and I_i counts those with two nonempty outputs. The smaller side is the minimum of the two output sizes, measured in sibling-list pair entries. Define $p_i = I_i/S_i$ for $S_i > 0$ and record NA otherwise. The two backends' logical traces agree. Counting one primary trace per input avoids multiplying events across backends, peer instances or replays. The parameter k in the targeted, balanced and skew construction is checked against the actual smaller side; repeated inputs are reported separately.

5.6. Records and Independent Validation

The preparation archive binds equivalence, qualification, calibration, source identities and environment. Each formal archive retains input and configuration identities, outputs, per-input power checks, execution order and individually committed compressed batches. Independent reconstruction

checks the first qualifying B, sample order, correctness and all reported medians, quartiles and ratios. The second attempt contains 12080 formal batches comprising 102080 checked sorting calls.

The preparation, formal and subsequent descriptive records are stored, respectively, in these repository directories:

```
docs/validation/unified_preflight_20261101/

docs/validation/unified_formal_20261101/

docs/validation/unified_followup_20261101/
```

Their original records remain unchanged. The formal archive contains both attempts, the restart decision, validation reports and analysis; only its complete second attempt supplies the main results. Source versions for the earlier studies remain indexed in chapter B.

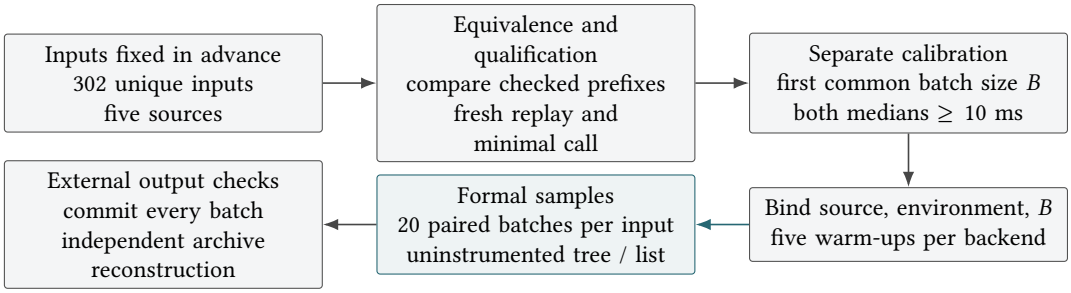


Figure 5.1.: Preparation and formal evidence flow for the unified five-source comparison. The same predetermined inputs and per-input B reach formal sampling. Checks and evidence persistence occur outside the timed intervals.

5.7. Historical Study: Separate Scope

The earlier rollback-enabled study compared Python sort, the oracle-backed reference pipeline and the ordinary-list core. Its recovery preparation, recognition scope, GC-disabled timing, interpreter and scheduling differ from the present experiment. Its records remain in the historical archive (section B.1). They are not pooled with the current comparison or used to infer a benefit from removing rollback.

6. Results

6.1. Overall Comparison

The uninstrumented tree is faster on 44 of the 302 measured inputs; the ordinary list is faster on the remaining 258. There are no exact ties. Each input contributes one ratio r_i of the tree's median time to the list's median time, as defined in section 5.5.

Giving every input equal weight, the median ratio is 1.257, with range 0.590–1.518. Figure 6.1 shows the spread across inputs. A further descriptive count places 14 inputs within $0.98 < r_i < 1.02$. These inputs are approximately tied under this chosen band, which was added after sampling. It leaves the strict $r_i < 1$ count above unchanged.

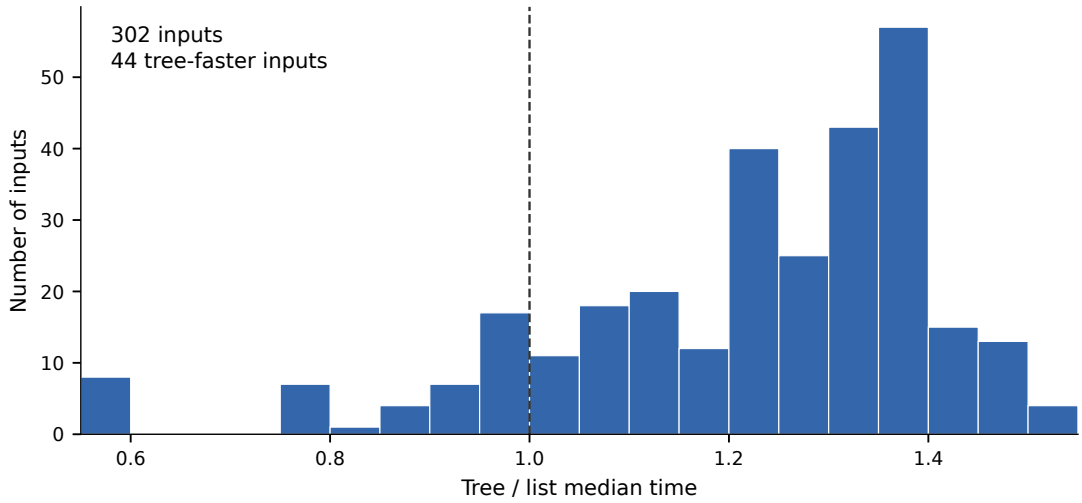


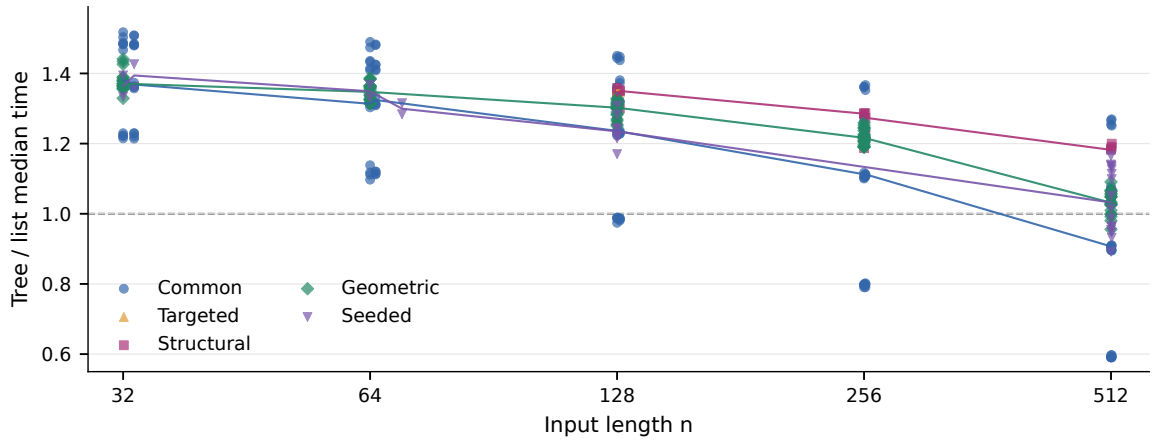
Figure 6.1.: Distribution of the 302 complete-session ratios, with bin width 0.05 and a reference at $r = 1$. This histogram is a supplementary descriptive figure added after sampling; it is not one of the four figures specified in advance.

6.2. Variation with Input Length

Table 6.1 groups the measured lengths into five bands. Their median ratios decrease from the smallest to the largest band. Tree-faster inputs first occur in the 128/129 band and are most numerous at 512/513. The bands have different source compositions, so their pooled medians do not isolate a size effect. Each of the four sources measured over several size bands (common, structural, geometric, seeded) has a lower median at its largest length than at its smallest. Figure 6.2 retains individual observations and the median at each exact length within each source.

Table 6.1.: Ratios by input-length band. Counts use unrounded ratios. The 64–70 band contains only the measured lengths 64, 65 and 70.

Length band	Inputs	Tree faster	Median	Range
32/33	52	0	1.373	1.214–1.518
64–70	52	0	1.335	1.097–1.490
128/129	74	8	1.304	0.974–1.450
256/257	54	8	1.215	0.790–1.367
512/513	70	28	1.037	0.590–1.269



Inputs at n	32	33	64	65	70	128	129	256	257	512	513
Common	14	14	14	14	0	14	14	14	14	14	14
Targeted	0	0	0	0	0	6	6	0	0	0	0
Structural	0	0	0	0	0	3	3	3	3	3	3
Geometric	20	0	20	0	0	20	0	20	0	20	0
Seeded	2	2	2	0	2	8	0	0	0	8	8

Figure 6.2.: Individual ratios and per-source medians at each exact length. Lines connect observed medians. The count panel gives the number of inputs behind every source/length median, replacing the overlapping labels in the archived figure. Observations are unchanged.

6.3. Variation with Structure

Structure is the clearest distinction within the common source. Table 6.2 compares its four predetermined categories. Each category has the same representation across lengths, although the strict-flat quota is smaller. The category medians decline from strict-flat to heavy nesting. No strict-flat or low-nesting input favors the tree, whereas tree-faster inputs occur in the medium and heavy categories.

Table 6.2.: Common-source nesting categories. S and I sum the performed splits and two-sided splits over one primary qualification trace per input; the backends agree on these counts.

Category	Inputs	Tree faster	Median	Range	S	I
Strict-flat	20	0	1.446	1.251–1.518	0	0
Low	40	0	1.355	1.050–1.488	844	0
Medium	40	8	1.231	0.895–1.382	2924	0
Heavy	40	24	0.986	0.590–1.230	20	0

The heavy category contains the strongest tree advantages. Among its 40 inputs, 24 favor the tree. Every heavy input has $S_i \leq 1$, and none has a two-sided split; 20 have no split at all. Thus the advantage on these constructions does not require frequent splitting. For the inputs without splits, it arises within the remaining work of complete sorting.

6.3.1. Structural Shapes

The structural source retains its balanced, skew and repeated shapes (table 6.3). None favors the tree. The repeated inputs have a lower descriptive median ratio than the other two shapes, but their complete operation histories differ. In particular, repeated inputs are a different construction from the single-interior-split balanced and skew inputs.

Table 6.3.: The three shapes within the structural source.

Shape	Inputs	Tree faster	Median	Range
Balanced	6	0	1.286	1.186–1.359
Skew	6	0	1.280	1.182–1.352
Repeated	6	0	1.205	1.041–1.299

6.4. Splitting and Complete-Sort Time

The targeted, balanced and skew inputs share the parameterized `values_for` construction. Within each measured n, their ratios vary little across the selected k values in figure 6.3(a). At the larger lengths, however, only the skew and balanced settings are available. The repeated construction has no single comparable k.

The fraction $p_i = I_i/S_i$ also shows no clear monotonic relationship with the ratio across this collection (figure 6.3(b)). An input with no performed split has $p_i = \mathrm{NA}$, not zero. The figure therefore omits 40 inputs from its horizontal axis while retaining them in all main timing summaries. Absolute split counts and smaller-side sizes are reported in chapter C.

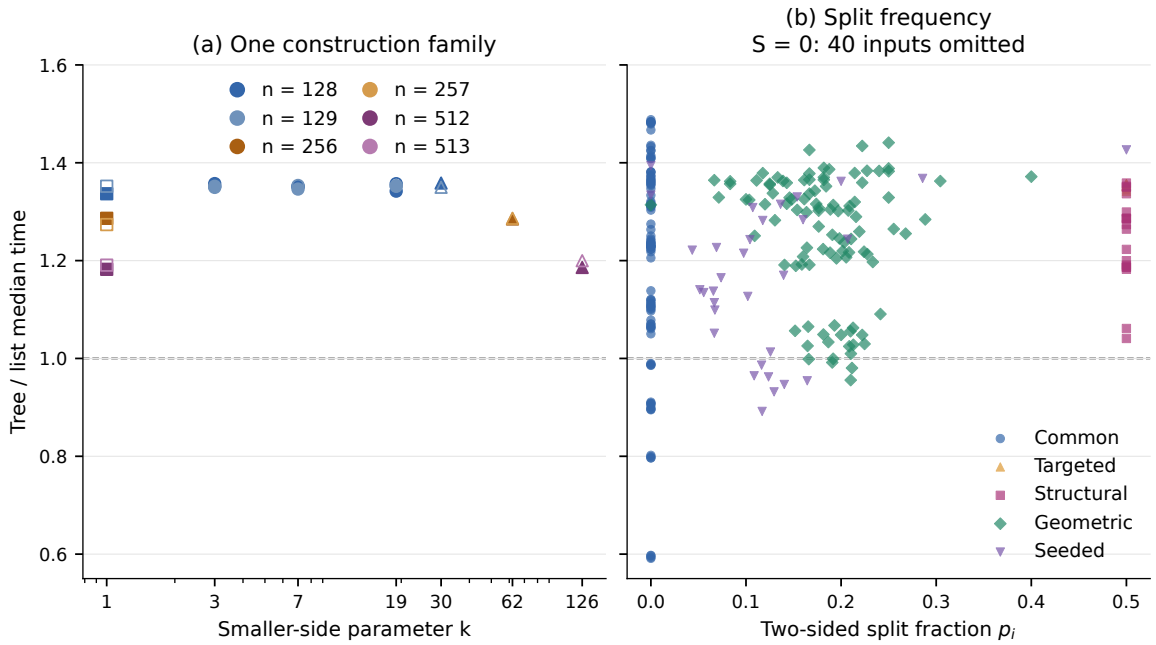


Figure 6.3.: Two predetermined descriptive plots combined into one figure. (a) The shared construction at each length: circles denote targeted inputs, triangles balanced inputs and squares skew inputs; open markers denote mirrors. Colors identify n. Repeated inputs are excluded. (b) One point per input with $S_i > 0$, colored and shaped by source.

6.5. Results by Source

All five sources completed qualification and formal sampling in the second attempt. Table 6.4 reports their registration, deduplication and completion counts alongside their timing summaries, as required

by the reporting protocol. Tree-faster inputs occur in common, geometric and seeded; none occurs in targeted or structural.

Table 6.4.: Complete reporting by source. Reg.: registered; Dup.: duplicates; Can.: canonical; Qual.: qualified; Timed: formally measured; Faster: tree faster. Targeted denotes `targeted_split`.

Source	Reg.	Dup.	Can.	Qual.	Timed	Faster	Median	Range
Common	140	0	140	140	140	32	1.231	0.590–1.518
Targeted	12	0	12	12	12	0	1.352	1.342–1.357
Structural	18	0	18	18	18	0	1.269	1.041–1.359
Geometric	100	0	100	100	100	5	1.302	0.956–1.441
Seeded	32	0	32	32	32	7	1.167	0.892–1.426

The predetermined all-input overview is retained in chapter C, together with the source-level split totals, smaller-side summaries and complete event-size frequency table. Those structural descriptions use the same qualified inputs as the timing tables. The underlying records and original four figures remain in the formal archive.

Summary of Observations. The tree is faster on 44 of 302 inputs, with a median tree/list ratio of 1.257. Median ratios fall across the measured size bands, and each of the four sources measured over several size bands (common, structural, geometric, seeded) has a lower median at its largest length than at its smallest. The tree's largest advantages occur on heavily nested inputs with almost no splitting. Neither the selected split parameter k at a fixed length nor the two-sided split fraction shows a clear relationship with the complete-sort ratio.

7. Limitations

7.1. Scope and Configuration

The results compare two concrete Python implementations: ordinary lists and the uninstrumented finger tree, both using the minimal sorting core. Source code, inputs and batch sizes stayed unchanged between preparation and the complete formal session. Earlier instrumented and rollback-enabled studies have different timing scopes and environments and are not pooled with these results. In the current implementation, an update failure may leave a partial state that must be discarded; invariants apply to completed updates.

7.2. Mathematical Model and Backend Fidelity

The physical and locality arguments have passed this project's source-level manual review. They have not been machine formalized or independently certified by a separate external reviewer. The endpoint and first-point rules are executable interpretations of details left implicit in the paper. The finite-parent locality case is an argument developed for this implementation; the published lemma covers the latest-child case. The mixed-operation potential argument also remains open to independent review.

The tree's linear total bound uses unit-cost comparisons and machine-word operations with expected dictionary lookup and amortized growth. Arbitrary Python comparison methods, large-number bit costs, interpreter scheduling and wall-clock behavior are outside it. Complete state audits and replay are also outside the minimal complexity claim. Changes to the implementation require reassessing the source-level argument as well as rerunning affected tests.

Ordinary lists do not inherit that bound: insertion copying alone is quadratic on increasing inputs, and their membership, ancestry and ownership work differs. The runtime comparison includes these differences in implementation. Timing curves alone establish neither complexity statement. The experiment's instrumented/uninstrumented equivalence check covers the 302 registered inputs; it is finite evidence, not a general proof about the source transformation.

7.3. Input Coverage and Scale

The 302 inputs come from five families chosen to vary nesting, sibling-list structure and interior splitting. They are controlled constructions, not a uniform sample from the space of valid Jordan sequences. Common uses fixed category quotas; targeted and part of structural share one parameterized construction. Geometric uses one grid-tree polygon generator, and seeded uses gap-selection heuristics. Mirrors, shared random streams and related parameter settings are dependent. The mixed median and faster-input count depend on this collection's composition.

The families leave structural gaps. Nesting depth and sibling-list length are not varied independently in a systematic design: a heavily nested suffix and an increasing flat construction test different operation histories. Balanced constructions deliberately include larger interior splits, while the geometric family's observed smaller side never exceeds 17 pair entries. The targeted and structural constructions provide only selected sizes and update histories, with parity and direction coupled in the structural source. They do not cover arbitrary combinations of deep nesting, long sibling lists and large repeated splits.

The geometric source contains only even lengths. A closed simple polygon in general position crosses a line an even number of times, and this generator uses only one sampled start and direction per polygon. Its fixed grid height also changes the aspect ratio as length grows. None of the inputs comes from an application dataset of real polygons undergoing clipping.

The largest measured length is 513; the 8 seeded inputs of length 1500 were excluded before sampling. Their earlier correctness records do not provide runtime evidence for this experiment. The lower ratios on larger and more heavily nested constructions describe these families over the measured range. They do not establish a crossover or continued improvement above that range, or the same behavior for other generators.

7.4. Timing Scopes and Environment

The timed call includes construction, the shared algorithm, output recovery and batching overhead. The backends retain different local checks, allocations and ownership work. The uninstrumented tree also binds methods when each instance is created. Complete-sort measurements do not isolate individual operation costs, including the cost of a split and the operations behind the tree's advantage on deeply nested inputs. Split fractions and sizes omit other parts of the work history, and operation counts describe the work mix without assigning runtime shares. The earlier instrumentation diagnostic likewise covers its selected inputs without assigning an additive share of total algorithm cost. The constructor microbenchmark describes its own fixed batch design, including automatic GC and

object retention; its comparison with the shortest sorting median provides a scale reference, not an isolated binding cost or a correction to sorting time.

The complete formal session uses one Apple M4 machine, macOS 27.2 and CPython 3.12.14 with AC power and low-power mode disabled. CPU affinity was not fixed, and actual core scheduling was not recorded. Warm-ups and alternating order do not establish stable frequency, temperature or absence of unrelated system activity. The restart is documented in section 5.4.2.

Sources ran in a fixed sequence. Session drift can therefore be confounded with source, while alternation balances only the backend order within an input. The prior cross-host diagnostic informed host selection; it does not make this single-host experiment a cross-platform performance study.

7.5. Statistical and Exploratory Reach

20 batches describe repeat variability for each fixed input/backend. Dividing duration by B does not yield B independent samples. Across inputs, median relative IQR is 0.965% for the list and 0.942% for the tree; the maxima are 2.601% and 3.080%, respectively. All batches remain included. The causes of the larger spreads were not measured, and variation cannot be assigned to CPU scheduling from these records.

Cross-input ranges and quartiles describe a collection of input summaries, not uncertainty intervals for a universal effect. The histogram and the 2% near-equality band are post hoc descriptive additions. The band is a descriptive threshold, not an equivalence test. The comparison of 170 overlapping inputs from the two attempts is also post hoc, outside the protocol and outside the main statistics; its observed changes do not establish equivalence. No significance tests, independent-sample confidence intervals or causal conclusions are drawn from these dependent constructions.

7.6. Output Provenance and Recognition

External certification and expected sorting support qualification and output checking for the paper cores. The reference pipeline instead returns oracle-derived output, as described in section 4.1.

All formal inputs are pre-certified valid. The evidence does not establish recognition accuracy, invalid-input failure reasons or recognition runtime. The sorting proof likewise starts from its validity precondition. Error correction and geometric applications remain outside this evaluation.

8. Conclusion

This thesis reconstructs Simplified Jordan Sorting as one executable core with ordinary-list and heterogeneous finger-tree backends. Both recover the sorted output from their maintained state. The implementation connects the paper's operations to a source-level cost argument and a reproducible runtime comparison.

8.1. Answers to the Research Questions

RQ1: reconstruction and correctness. The shared Step 1/2/3 loop maintains geometric ownership, exact sibling groups and sorted prefixes. The argument in chapter A explains the two odd-stage z_1 corrections, temporary transfer, reversed ribs and recursive repair transients. Stable pair identities and child groups survive splitting. All 302 inputs passed the experiment's equivalence and qualification checks, and every saved output in the 12080 formal sample batches of the complete attempt agrees with its expected sorted order. These checks support the reconstruction on the tested inputs without replacing its argument.

RQ2: observed runtime. The uninstrumented tree is faster on 44 of 302 inputs; the ordinary list is faster on the remainder. The input-level tree/list ratio has median 1.257 and range 0.590–1.518. The pooled median decreases across the measured size bands, and the common source's more heavily nested categories have lower ratios.

RQ3: structure and cost. One potential pays propagation across mixed updates, giving amortized constant endpoint insertion and smaller-side logarithmic split. A list-size potential then pays the split terms over the full run, including bounded ownership and shared-core work. The linear total uses unit-cost comparisons and machine-word operations, expected dictionary lookup and amortized growth. Ordinary-list insertion copying alone has a quadratic lower bound.

The five sources cover different splitting histories, including two-sided splits in targeted, structural, geometric and seeded inputs. Yet the strongest tree advantages occur among heavily nested common inputs with at most one split and no two-sided split. Within the shared targeted/balanced/skew construction, ratios vary little with the selected smaller-side parameter at a fixed length.

8.2. Contributions and Further Work

The thesis provides an implementation whose operations can be traced to the reconstruction and cost argument, together with reproducible measurements. Several extensions would test how far the observed trends carry. Generators that combine depth with long sibling lists, and application-derived inputs with large interior splits, would address structural gaps in the current families. Open polygonal chains would add odd lengths, while multiple starts and directions per polygon would test dependence on traversal choice.

Larger inputs, including the excluded length-1500 cases, would test whether the narrowing of the tree/list ratio continues beyond $n = 513$. Intersection sequences from real polygon data would connect the measurements to the clipping application of Fung et al. Operation counts from the instrumented tree could describe differences in work mix and guide targeted measurements of candidate explanations. Independent review of the finite-parent locality extension and the mixed-operation potential argument would strengthen confidence in the cost analysis. Recognition, invalid-input behavior and error correction each require their own specification and evaluation.

A. Representation, Locality and Cost Argument

This appendix develops the manual source-level argument for the measured backend.[1] The correspondence between proof steps and source code is indexed in chapter B. The geometric precondition is a simple curve with distinct, totally ordered intersections. The final cost statement uses unit-cost comparisons, machine-word operations and constant-size allocation, with expected dictionary access and amortized growth. Checked audits, replay and input certification are excluded.

A.1. Decoding and Preservation

Definition A.1 (Complete representation D)**.** Decode each header as an ordered external red–black tree. External leaves are black with rank zero and span one. An internal separator is the maximum left key; its children have equal rank, and its rank is that common rank plus one exactly when it is black. No red node has a red child, and the complete root is black. Internal span means full leaf count in an interior node, inward-branch count on a rib, and zero at the root. Header size is the total leaf count, including for a singleton.

On a left rib the left slot points up; on a right rib the right slot points up. The other slot points inward. Decoding reverses these outer chains; off-rib children already point down. Internal-root slots are empty, and header top caches identify its two logical children. Fingers identify extreme leaves; decoding is acyclic and reaches every node once. Stable leaf/handle identities bind pair records. Only endpoints carry boundary-header references. Each live header has one owner and occurs in that owner's ordered slots, of which there are at most two.

Lemma A.2 (Conversion and endpoint insertion)**.** *Conversion between a downward component and its finger representation preserves its decoded leaf sequence and identities. Singleton creation and legal endpoint insertion restore D without scanning inward subtrees.*

[1] The English and LaTeX exposition of this chapter was prepared with Codex from my extensions of the arguments of Fung et al. and revised during the review process; see Section B.2.

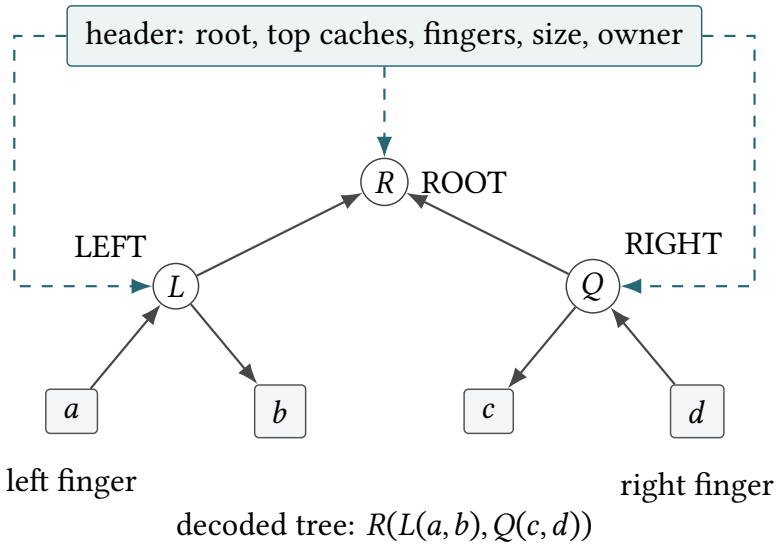


Figure A.1.: Physical decoding: solid arrows are node links; dashed arrows are header references. Outer chains point up and inward branches down; root slots are empty. Header fingers reference leaves a, d (arrows omitted). Colours and sizes are omitted.

Proof. When exposing a rib, save its outward child before replacing that slot by the upward link. The processed chain and untouched remainder still decode to the original tree; each new rib span is the unchanged inward count. The inverse saves the upward link, attaches the already processed lower component and refreshes rank and full span bottom-up. A leaf needs no conversion; a nonroot top exposes one rib, an internal root two. Final root blackening may increase rank by one.

Insertion attaches a black leaf and red connector at the selected finger. Only the connector's upper edge can violate red adjacency. A red uncle is blackened with the parent while the grandparent becomes red; the latter's rank is unchanged and the conflict moves upward. With a black uncle the left-side rotation is $G(P(C, M), F) \mapsto P(C, G(M, F))$. Initially P, C are red and G, F black. The three blocks C, M, F have equal rank. Afterwards G is red, P black, and the total rank is unchanged. Separators remain $\max M$ at G and $\max C$ at P.

For nonroot G, it becomes interior with children M, F; P retains its rib role, points to the former ancestor and has inward child G. For root G, P becomes the empty-slot root, G becomes the opposite rib with inward child M, and top caches become C, G. The unchanged upward links from C and F already name their correct parents. Spans follow these roles. The mirror is $G(F, P(M, C)) \mapsto P(G(F, M), C)$, with separators $\max F$ and $\max M$, still maxima of left blocks. Black-parent termination, rotation or root blackening restores D. Only the replaced endpoint's boundary slot changes; the live list ID is retained. □

Lemma A.3 (Recursive downward join contract)**.** *Joining two disjoint ordered components preserves their leaf union and order. The recursive return has equal child black heights and correct metadata, but may contain one root–child red edge when its root is red. A complete nonempty join restores the red–black invariants.*

Proof. Both input roots are blackened. Descent follows the right path of a taller left component, or the left path of a taller right component; the other argument remains the original blackened root. A black step lowers rank by one and a red step preserves it, so descent reaches equal-rank black roots without crossing their rank or changing sides. There a red connector forms an internally valid base case with the same rank. On return, the replaced component keeps that rank. A black parent without conflict is refreshed; only an outer RR conflict on right descent, or LL on left descent, can arise. One rotation produces a red middle root with two black children and unchanged black height. The LR/RL helper branches are defensive: direct helper fixtures exercise them, but valid recursive joins do not reach them.

A red parent takes the direct-return branch. Its originally black updated child can return an internally valid red root; the other child stays black. Consequently only the updated root–child edge may be red–red. This is a recursive transient, not D. The original parent of that red node is black and repairs this LL or RR conflict on the next return. The induction excludes a deeper or second simultaneous red conflict. Descent never searches through a return transient.

At the top, either the valid base case or a black-root repair returns an internally valid tree. Final root blackening changes only root colour; it cannot repair an internal red edge. For example, joining $A_B([0], P_R([1], [2]))$ to $[3]$ temporarily returns $P_R([1], C_R([2], [3]))$, then repairs at A_B; the mirror uses LL. In a left rotation $A(X, B(Y, Z)) \mapsto B(A(X, Y), Z)$, separators stay $\max X, \max Y$; refreshing bottom-up gives full spans and ranks. The connector separator is the left input's maximum and sizes add. An empty-input branch simply returns the other component and retires the unused connector; this branch alone need not blacken its root. □

Lemma A.4 (Detachment and split preservation)**.** *Legal split and adoption preserve the leaf set, stable identities and descendant groups, and restore D and header ownership at their completion.*

Proof. Detachment removes an extreme component and its parent connector, replacing the latter by its sibling. A red parent loses no black height; a red sibling replacing a black parent is blackened. Otherwise the replacement is short by one black layer. Use a transient D^*: the short component's rank is actual, but ancestors still account for one virtual black layer on that path. Refreshing these ancestors as ordinary equal-height trees would be invalid.

For short rank r, the sibling has rank $r + 1$, so is not a leaf. A red sibling is blackened and the black parent reddened before rotation, reducing to a black sibling. If both nephews are black, redden the sibling: a red parent absorbs the deficit by blackening; a black parent decreases rank and becomes the next short component. A red near nephew with black far nephew is rotated and recoloured into the far-red case. In that case the sibling takes the parent colour, the parent and far nephew become black, and a rotation absorbs the deficit. A deficit reaching the root lowers the whole black height by one. These are all repair branches.

For a left deficit the outer rotation is $P(S, B(N, F)) \mapsto B(P(S, N), F)$, with separators $\max S, \max N$. P is a left rib pointing up to B and inward to N, with span $|N|$. At the root, B gets empty slots, span zero and top caches P, F; otherwise it retains the rib with inward F and the old upward link. The mirror keeps separators $\max F, \max N$ in $P(B(F, N), S) \mapsto B(F, P(N, S))$. A near rotation at the opposite root rib changes $B(N(X, Y), F)$ to $N(X, B(Y, F))$: the inward spans become $|X|, |Y|$, and the appropriate top cache becomes N. The far upward link is unchanged. These substitutions preserve order and update only affected metadata.

The selected component need not straddle the cut: it can lie wholly on one side. The remainder lies wholly on a known output side. Downward splitting classifies leaves by key $\leq v$ or $> v$; each saved sibling is joined once to the corresponding accumulator in order, using lemma A.3. Connectors are reused once or retired, never leaves. Finger joins stop at equal black rank and use lemma A.2; conversion handles their component boundaries. Size starts at one at a finger and accumulates inward spans on ascent; at the root use header size. The remainder is obtained by subtraction, downward sizes by full spans, and joined sizes by addition. There is no output-wide counting pass.

Every split retires the source header, including an empty-side fast return, and allocates fresh IDs for nonempty outputs. It clears old endpoint references and sets new ones. Adoption checks the pending source, owner, new parent and output IDs, then changes headers and at most two child slots. Leaves, handles and their own child groups are unchanged. Strict containment of extreme children implies containment of the whole ordered group. In the sorting call the new parent was childless, so acquiring its strictly contained old children cannot create a cycle. Pending overlap is allowed only for this planned transfer; completion clears it before output insertion. □

A.2. Locality and Exact Grouping

Use G from section 3.2.1, not merely a bound of two groups. An owner is the smallest strictly enclosing same-parity pair, or the parity dummy. Its region R is bounded by its arc and its direct

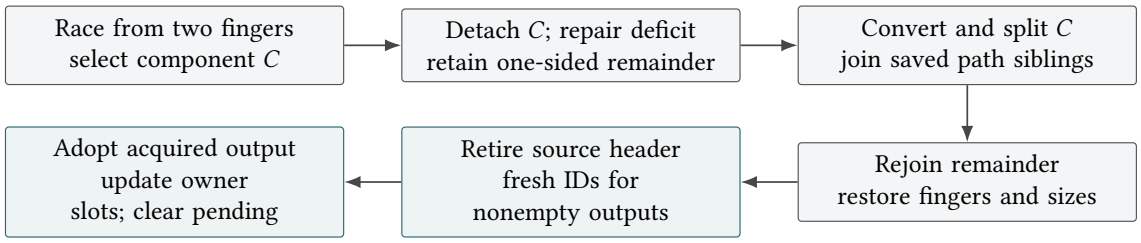


Figure A.2.: Two-sided split and adoption. The selected component may itself yield an empty side. Sizes come from path spans and header totals; transfer changes header ownership rather than rebinding every leaf.

children's arcs, with intervening open axis gaps. Descendant arcs lie behind child boundaries, not inside R.

Lemma A.5 (Applicable re-entry gaps)**.** *Suppose the region has not changed since its latest member P_j was added. If P_j is a child, first re-entry from z_j uses its adjacent region gap, or, for the dummy, the opposite unbounded gap. In the latter case, z_j and the re-entry point z_k bracket all other finite endpoints of the region. If P_j is the finite parent, first re-entry uses the gap immediately inward from z_j.*

Proof. The child case is the locality lemma of Fung et al. [2, p. 88, locality lemma (a)–(b), Figs. 3–4], under its latest-child and unchanged-region hypotheses; part (b) supplies the dummy bracketing condition. It is not applied to a latest finite parent. For that additional case let q be the first child endpoint inward from z_j, or the other parent endpoint if there is no child.

Assume the first re-entry z_k is outside (z_j, q). Choose a child endpoint a strictly between z_j and z_k along that boundary, and let $b = z_{j-1}$ be the other parent endpoint. An auxiliary simple crosscut inside R connects z_j to z_k and separates these boundary vertices. The actual curve portion from z_j to z_k lies outside R by first re-entry and meets the crosscut only at its ends. Together they form a simple closed curve separating a and b. The old curve portion connecting a to b cannot cross the crosscut, because the old curve avoids the region's interior; it cannot meet its newer endpoints or cross the actual later portion, by simplicity. This contradicts separation. If there is no child, the entire inward gap is already (z_j, q). Unlike the dummy, a finite parent has no second unbounded outside gap. This establishes the extra case. □

Lemma A.6 (Legal calls and prefix induction)**.** *Starting from initialization, G, geometric ownership and the sorted-prefix invariant imply legal endpoint insertion, same-owner/different-group transfer when required, and correct output insertion. Each completed iteration restores these invariants.*

Proof. Initialization has one singleton per parity family and correctly orders three points. Consider iteration i, with $x = z_{i-1}$ and new point $y = z_i$. Let p be the smallest same-parity interval containing x, or the dummy, and let P_j be the latest member of its owner region. Updates inside older descendants have not changed that region. Were there an earlier first re-entry, its next same-parity arc would create a new direct child of p, possibly acquiring consecutive children. Its end index would exceed j, contradicting the definition of j. Here alternation between the two parities is essential. Thus lemma A.5 applies to x.

Except for x and, at odd i, z_1, processed points are endpoints of same-parity pairs. Step 1/2 skips this exceptional z_1. Parent and direct child endpoints block all more distant ancestors and descendants in the neighbor search. Consequently an interior entry gap gives the last endpoint of the left group and first endpoint of the right group; an outer gap gives one group boundary and the parent. By G these are designated endpoints, not arbitrary members. The two groups, when both present, have the same owner and different live IDs.

For $x < y$, simplicity implies that y remains inside finite p and is neither inside a direct child's interval nor equal to its endpoints. If the left boundary is p, Step 3(a) creates a singleton in a free slot. In the dummy outer-gap case, the bracketing condition leaves the group beyond z_j empty, so the owner has at most one group. When the left boundary is a child, Step 3(a) instead appends to the left group. If the right boundary is p, there are no children to transfer. Otherwise its right group is the source. Reading that source before insertion is safe: Step 3(a) changes only the other group or a new slot, preserving source identity and owner.

All source children lie to the right of x. Since y cannot lie inside a child, a source key $\ell_h \le y$ is equivalent to $x < \ell_h < r_h < y$. Hence value splitting acquires precisely the consecutive children enclosed by the new pair. The temporary overlap is exactly the declared transfer. After adoption, p's latest child is P_i and its groups are separated by y; P_i is the latest parent of its own single acquired group. Other owners' groups, descendant identities and the opposite parity are unchanged. Strict enclosure gives acyclicity.

Among old paired endpoints between x and y, the largest is the high endpoint of the rightmost acquired direct child, or x if none exists. Descendants cannot exceed that child's extreme. This is the predecessor ignoring the exceptional odd-stage z_1; replacing the anchor by z_1 exactly when it lies between this base and y gives the true predecessor. The decreasing case reflects the argument, using the low endpoint and successor. Step 3(c) therefore preserves sorted order after the pending-state fence. This proves the simultaneous prefix induction. □

A.3. Costs of Mixed Operations

Lemma A.7 (Cumulative join cost)**.** *Splitting a downward component of raw rank r costs $O(r+1)$ in joins and path work. It does not require $O(r)$ joins each independently charged $O(r)$.*

Proof. Write v for rank after blackening the root. For nonempty join inputs of normalized heights a, b, the result has height c and descent count Q with

$$\max(a,b) \le c \le \max(a,b)+1, \qquad Q \le 2|a-b|+3.$$

Descent follows a legal tree: each black level decreases the height difference and there cannot be consecutive red nodes. Each return repair is constant work by lemma A.3. An increase in height leaves a black root with black children: before final blackening it was the valid red base or a black-root repair result, not the red-root transient. Pre-blackening an originally red input adds at most one separately.

Inductively, splitting a legal raw-rank-r component gives nonempty outputs of normalized height at most $r+1$. For a black input attaining $r+1$, the output root and children are black. For the induction step let child rank be q. At a black root, both join inputs have normalized height at most $q+1$, so the output is at most $q+2 = r+1$, with the stated strong case. At a red root the children are black. If the recursive output reaches $q+1$, its root and children are black by that strong case; joining a height-q sibling descends one level and cannot increase height again. Otherwise the ordinary join bound suffices.

Thus at each backtracking join, accumulated height a and untouched sibling height b satisfy $a-b \le 1$. The displayed bounds imply $Q \le 2(c-a)+5$. Each of the two output accumulators has nondecreasing normalized height, bounded by $r+1$. Summing over $J = O(r+1)$ joins gives $\sum Q \le 2(v_L+v_R)+5J = O(r+1)$. Empty-side returns add only constant work per frame. This pays descent, reconstruction and metadata maintenance. □

Lemma A.8 (Finger distance and opposite-rib work)**.** *For smaller nonempty output size s, the selected component has rank $r = O(\log(s+1))$. Opposite-rib access in detachment costs $O(r+D+1)$, where D counts continuing deletion propagations.*

Proof. The left search climbs only while the parent separator is below the cut; the right search has the reflected condition. It is the component *before* a climb that is known to be wholly on one side, not necessarily its parent. A one-sided component of rank t has at least 2^t leaves. On the smaller side this is at most s, and one climb increases rank by at most one. Alternating searches differ by at

most one step; no two red levels are consecutive. Hence both paths, and the selected component, have the stated rank bound. If $s = 0$, the endpoint fast path costs $O(1)$.

An inward sibling is interior except at the root. Only there can `rib_child` require an opposite-rib walk. A red-sibling root rotation makes the continuing parent nonroot; propagation at the root terminates. A near rotation permits at most one additional far-nephew lookup. There are therefore at most two such walks per repair. With initial short rank $r_0 \le r + O(1)$, each continuing propagation raises actual rank by one; the reached root has rank at most $r_0 + D + 2$. This proves the walk bound. Conversion visits $O(r + 1)$ levels; finger joining climbs the large component only to the small component's rank, also $O(r + 1)$. Any ensuing insertion propagation is charged next. □

Lemma A.9 (One potential across mixed updates)**.** *Endpoint insertion costs amortized $O(1)$, and split costs $O(\log(s + 2))$, including local repairs, conversion and size maintenance. Creation and adoption have constant cost.*

Proof. On decoded logical internal nodes define

$$\Phi = \#\{\text{black nodes with two black children}\} + 2\#\{\text{black nodes with two red children}\}.$$

Count each node once across the live forest and temporarily disjoint components; never reset the potential when headers change. Leaves contribute zero. An insertion propagation changes a black/red-red grandparent of weight two into a red node of weight zero; the blackened parent is mixed and the blackened uncle has two black children, contributing one. The decrease is one per continuing propagation. An upper boundary or terminal repair changes only constantly many weights.

In a continuing deletion propagation, the black sibling with black children becomes red, losing one, and the black parent changes from two black children to mixed, losing another. The next ancestor's child stays black. Thus each such step decreases Φ by two; absorption, rotations and root treatment change constantly many weights. Direction conversion alone changes neither logical children nor colours. The $O(r + 1)$ join/path steps each affect a constant number of weights, and any finger-join propagation uses this same potential. For constants independent of the sequence,

$$\Delta\Phi \le C(r + 1) - D - I, \qquad W \le C'(r + D + I + 1),$$

where I counts propagation rounds, not individual recolourings. Choose a sufficiently large constant multiplier of Φ to pay both propagation and the D term in the opposite-rib work. Lemmas A.7 and A.8 then give the split bound. The same accounting with constant boundary work gives endpoint insertion. Adoption touches bounded header/owner slots, not a leafwise ownership map. □

Theorem A.10 (Total valid-input work)**.** *Under the stated computation model, the shared minimal core with this finger backend uses $O(n)$ expected/amortized work and $O(n)$ space on a valid input of length n.*

Proof. By lemma A.6, each iteration performs constantly many legal boundary accesses and at most one creation/insertion, split and adoption. To pay the split terms use the nonnegative list potential $\Psi = c \sum_{L \neq \emptyset}(|L| - \log_2 |L|)$. Creation increases it by c; insertion into size m increases it by $c(1 - \log_2(1 + 1/m)) \leq c$; adoption changes no sizes. For a nonempty split $m = a + b$ with $1 \leq a \leq b$,

$$\Delta\Psi = c \log_2 \frac{a+b}{ab} \leq c(1 - \log_2 a).$$

Since $\log_2(a+2) \leq \log_2 a + \log_2 3$, a sufficiently large c pays the smaller-side term. An empty-side split has $\Delta\Psi = 0$ and constant amortized cost. Combine this with the single multiple of Φ in lemma A.9; initialization has constant potential.

The remaining shared-core work consists of bounded endpoint/handle and owner-slot accesses, geometric comparisons, stage records and linked output insertion per iteration. It performs no membership or ancestor scan in this backend. Input materialization and final output traversal are linear. Access wrappers and iterator resumption add constant work per affected access, not zero work. Constant-size records per iteration and linear node/handle registries give the space bound, including retired-header bookkeeping. Nonnegative final potentials complete the aggregate bound. □

The theorem accounts for the shared minimal core and its tree operations in the model stated at the start of this appendix. The measured runtimes are reported in chapter 6.

B. Evidence and AI Use

B.1. Evidence and Source Index

Current experiments. The unified comparison uses 302 inputs and the source binding `bca4c1c6`. Both formal attempts ran from the clean checkout `2ae92952`, whose Python sources retain that binding. Only the complete second attempt supplies the main results. The input registry is under `docs/validation/input_suite_registry_20261101/`. Source versions below identify execution code; archive commits identify the later storage of the records.

Unified preparation. Equivalence, qualification and calibration ran at `bca4c1c6`; the archive was committed as `d0d47a67`. Its root is

```
docs/validation/unified_preflight_20261101/
```

The `equivalence/`, `qualification/` and `calibration/` subdirectories retain the original records. The `checks/` directory contains the independent validation reports before and after relocation. `freeze.json` and `production_sha256.json` bind the source files.

Formal attempts, validation and analysis. The records were archived at `1f1d68e5` under

```
docs/validation/unified_formal_20261101/
```

The following paths are relative to that root:

- First attempt: `unified_formal_bca4c1c6/`.
- Complete second attempt: `unified_formal_bca4c1c6_restart_20261101_01/`.
- Analysis: `unified_analysis_bca4c1c6_restart_20261101_01/`, including `cases.json`, `summary.json` and the four original figures.
- Independent validation after archival: `checks/archived_first_attempt_validation.json` and `checks/archived_second_attempt_validation.json`.

Each attempt also has a sibling `.validation.json` file using its run-directory basename. The diagnosis, restart decision and author review receipt remain beside the runs; the README indexes their exact filenames. The restart is described in section 5.4.2.

Constructor benchmark and stability check. The follow-up plan, scripts and execution version are `70d47e91`; results were archived at `7006629f`. The production source binding remains `bca4c1c6`. The archive root is

```
docs/validation/unified_followup_20261101/
```

The constructor configuration, environment and raw batches are under `results/constructor/`. The analysis directory `results/analysis/` contains `constructor_summary.json`, `stability_summary.json` and the `stability_cases.csv` and `stability_cases.json` records for the 170 overlapping inputs. Independent checks are saved as `results/verification.json` and `checks/archived_verification.json`. The stability comparison is a post hoc supplement outside the main statistics. Each archive's README records its path bindings and read-only validation commands.

Earlier instrumented experiments. The measured version is `4a83fa6` (5,600 common-set and 480 targeted-set batches). Its Python sources are identical to those of the calibration version, `c5b5ac3`. Qualification sources are `3cafb44` for the 140-input common set and `668371f` for the 12-input split set.

Earlier instrumented structural extension. The eighteen-input November 1 campaign uses `bdfcdcf8` for qualification, calibration and formal sampling (720 sample batches). All production sorting sources match the original formal version. The exact input selection, protocol, raw records and reconstructed results are under `docs/validation/structural_split_formal_20261101/`. This campaign retains the existing review receipt for unchanged production code; it does not claim a new independent proof review.

Development evidence. Source identities for table 4.1 are `cc1aa2d` (operation tests), `84f7e13` (structural campaign) and `19a553fb` (seeded supplement, October 20 protocol). The core access interfaces were introduced at `71e3e71`.

The archive index and proof-to-source map cover development, historical and supplementary evidence: `docs/thesis/evidence_source_index_20261022.md`.

B.2. Documentation of AI Use

Tools used

Tool	Provider / model	Period of use
ChatGPT	OpenAI, ChatGPT 6 Sol	June and September 2026
Codex	OpenAI, GPT-5.6 Sol; including the `academic-humanizer` skill, version 0.3.3	September–November 2026
Claude	Anthropic, Opus 5.5, medium setting	October–November 2026

Influence on process and product

Phase	Tool	Use	Affected parts
Understanding and planning	ChatGPT, Codex	Translation, algorithm explanations, examples of sibling groups and finger trees, literature searches, architecture discussions and discussion of review findings	Research process; background and implementation explanations
Implementation	Codex	I wrote the shared sorting core (Steps 1–3) and the finger-tree backend. Codex wrote the tests and the validation and experiment tooling, assisted with later revisions of the core and backend, and wrote the module that derives the uninstrumented tree from the backend source.	Test files; validation and experiment scripts; revisions in `src/`; `src/uninstrumented_finger.py`
Proof	Codex	English and LaTeX formulation of my extensions of the arguments of Fung et al., source checking and subsequent revisions	Appendix A
Experiments and results	Codex	Calibration and measurement tooling, inspection of saved records, reconstruction of result tables and revision of their interpretation. On 1 November: implementation and equivalence validation of the uninstrumented timing configuration; the unified input registry; execution of preparation (equivalence, qualification and calibration); validation and analysis of formal results; the constructor microbenchmark and the stability check.	Experimental pipeline and archives; Chapters 5 and 6 and related summaries

Continued on the next page

Phase	Tool	Use	Affected parts
Writing	Codex	English and LaTeX drafting and revision; use of Academic Humanizer to reduce repetitive qualifications and improve clarity	Abstract, main chapters and appendices
Review and planning	Claude	Technical and manuscript review (omitted proof premise, incomplete diagnostic table); review of wording and terminology, with suggested wording for the abstract, research questions and selected definitions; discussion of the experimental design, including the geometric input probe, the input suite and the analysis plan; drafting of instructions for Codex	Appendix A; abstract; Chapters 1, 5 and 6

All measurement values reported in Chapter 6 come from the archived program runs indexed in Section B.1. Codex assisted with the tooling used to check and present those results.

On 1 November 2026, I proposed using the uninstrumented configuration for formal timing. Claude participated in the experimental-design discussion and drafted instructions for Codex. I initiated formal sampling on my local machine, with Codex running the commands at my explicit request. Codex wrote the author review receipt only after I confirmed that I had completed the review and explicitly instructed it to create the receipt.

The following examples contain excerpts of recorded requests. English prompts are translations of the Chinese originals; descriptions of outputs are summaries rather than verbatim transcripts.

Example 1: reducing repetitive qualifications

Claude review, revision carried out with Codex, 31 October 2026

Prompt excerpt:

> "Do you think the repeated emphasis on the limits of the evidence needs changing?"

Output, summarized:

Claude's review identified repeated qualifications that obscured the findings. Codex revised the text accordingly, using the Academic Humanizer skill to improve wording and transitions while retaining the relevant assumptions and scope.

Use:

The revisions were incorporated into the manuscript. Checks for this revision compared equations, mathematical environments, table contents and citations with the preceding version and confirmed that these remained unchanged.

Example 2: correcting a proof premise

Claude review, revision carried out with Codex, 20 October 2026

Request excerpts forwarded to Codex:

> "Add the missing premises to A.5/A.6."
>
> "Ensure that the bracketing condition appears explicitly in the text, and identify its source as Fung, p. 88(b)."

Output, summarized:

Codex checked the relevant passage in the source paper, added the bracketing premise to Lemma A.5 and explained its role in the available group slot in Lemma A.6.

Use:

I decided to address this finding and specified the required change and its source in the requests above. The correction was incorporated into Appendix A. It changed the proof exposition; the sorting implementation was unchanged.

Example 3: completing the diagnostic table

Claude review, revision carried out with Codex, 20 October 2026

Request excerpt forwarded to Codex:

> "Show all fifteen rows in Table 6.5, or present the fifteen combinations of input size and nesting level."

Output, summarized:

Codex reconstructed the complete table from the saved measurements and revised the accompanying interpretation. At that stage, the table reported four cases favoring the tree with field-access instrumentation removed and eleven favoring the list backend.

Use:

I decided that the table should report all fifteen cases and specified the required format. At that time, the complete table was added to the results chapter, and its interpretation was reflected in the abstract and conclusion. No new timing measurements were taken for that revision. When the November experiment switched to uninstrumented timing, the diagnostic table was removed from the results chapter. In the main text, the diagnostic findings are now mentioned briefly in the "Configuration choice" paragraph of section 5.3.

B.3. Reflection on AI Use

AI tools supported my work on an executable reconstruction of Jordan sorting. I required the sorting procedure to recover its output from its own maintained state, and I distinguished correctness, complexity arguments and measured runtime when assessing revisions. These requirements shaped how I wanted the reconstruction and its results to be evaluated. My supervisor suggested attempting an efficient backend. After studying Fung et al., I chose a heterogeneous red–black finger tree for it. I wrote the shared sorting core and the tree backend; Codex wrote the tests and the experiment tooling and assisted with later revisions of my code.

During revision, I questioned whether the common set's lack of two-sided splits left a gap in experimental coverage. The resulting discussion distinguished twelve targeted inputs with formal timings from eighteen structural inputs that had at that stage undergone correctness checks without formal timing. This clarified an important limitation: variation in nesting depth does not by itself establish coverage of the operations that distinguish the backends. It also helped me formulate a more specific question about which additional workloads would strengthen the comparison. Those eighteen inputs were subsequently timed in a separate instrumented extension and then included in the unified comparison reported in section 6.3.1.

ChatGPT helped with translation, algorithm explanations and discussions of review findings. Concrete examples were useful when I was trying to understand the relationship between sibling groups and the red–black finger tree. Codex assisted with implementation changes, tests, validation tooling and manuscript revisions. I found literature searches, architecture discussions and help expressing technical ideas in English particularly useful.

For Appendix A, Codex helped formulate my extensions of the arguments of Fung et al. in English and LaTeX. Claude subsequently identified an omitted premise and an incomplete diagnostic table. I decided to address both findings and specified the required changes; Codex checked the premise against the source paper and carried out the revisions. The omission showed me that readable mathematical prose can still leave an argument incomplete. Successful tests and fluent explanations therefore required different kinds of scrutiny.

I also questioned repeated qualifications that obscured the main findings and requested a more direct presentation. For small edits whose intended wording was already clear, working directly on the text might have been faster. When revising the introduction, I supplied replacement wording to remove a repeated description of the two backends. Explaining such a local change and checking another version offered limited additional benefit.

I noticed that timing included the overhead of operation-counting instrumentation and proposed using the uninstrumented configuration. I am satisfied that both backends produced correct output in the formal measurements. With instrumentation removed, the tree was faster on 44 of 302 inputs, and the gap narrowed across the measured size bands; the thesis reports these results alongside the tree's stronger structural bound. I am less satisfied with the incomplete documentation of some earlier decisions and AI assistance. An independent human review of the locality and mixed-operation arguments would also strengthen my confidence in the work. In future projects, I would record decisions and their reasons as they arise. Responsibility for understanding and defending the submitted work remains mine.

C. Supplementary Experimental Displays

This appendix retains the predetermined all-input overview and the full source-level structural descriptions. The main results contain the other three predetermined plots: ratios by length and the two split-related panels. Only the main-text histogram is an additional plot. All observations come from the same complete formal attempt and its qualification evidence.

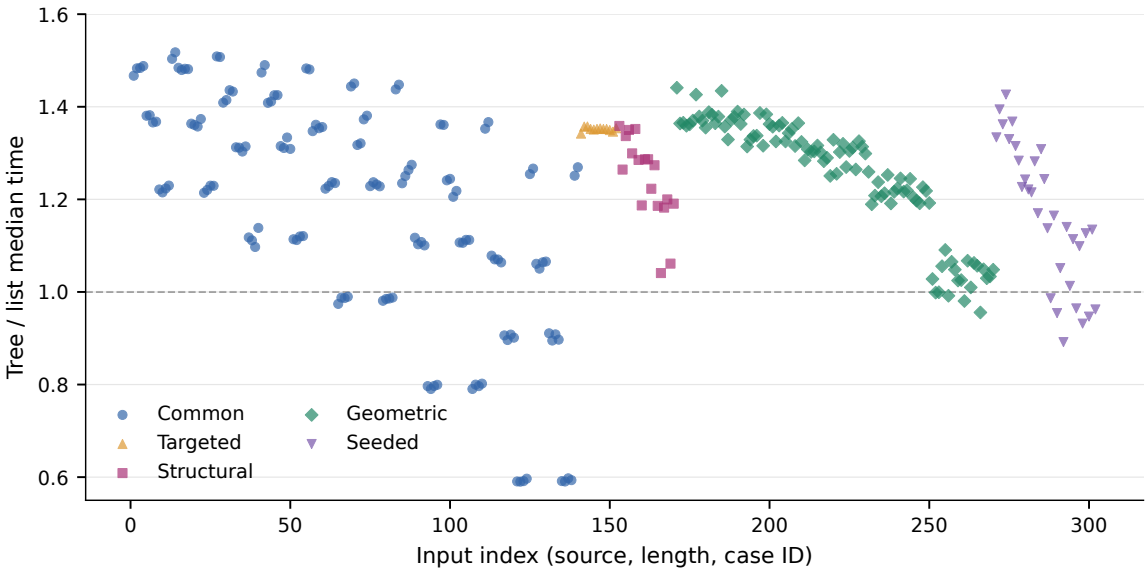


Figure C.1.: A paper-sized layout of the predetermined overview, with unchanged observations and ordering. The plan registers 302 inputs; deduplication retains 302, and all 302 are measured in the complete attempt. The horizontal order is source, length and case identifier, not performance. The exact order is available in the archived analysis.

C.1. Qualification Structure by Source

Table C.1 reports one primary trace per input. Logical split counts agree between the backends; peers, replays and the second backend are not counted again. S and I are the sums of performed splits and

two-sided splits. N_I counts inputs with at least one two-sided split, and N_0 counts inputs with no performed split. The smaller-side statistics are over two-sided split events and use pair entries as their unit. Their event denominator differs from the input denominator used for runtime ratios.

Table C.1.: Structural descriptions of all qualified inputs. The final column gives the minimum, median and maximum smaller-side size. NA denotes an empty event distribution.

Source	Inputs	S	I	N_I	N_0	Min./med./max.
Common	140	3788	0	0	40	NA
Targeted	12	24	12	12	0	3/7/19
Structural	18	460	230	18	0	1/1/126
Geometric	100	9272	1774	99	0	1/1/17
Seeded	32	2946	307	30	0	1/1/8

The complete smaller-side frequency distribution is given in table C.2. Common has no two-sided splits, so its distribution is empty; zeros in that column indicate absence of events, not zero-sized smaller sides. No split events are treated as independent runtime samples. The archived per-input table also retains every S_i, I_i, smaller-side event list and p_i, including NA when $S_i = 0$.

Table C.2.: Number of two-sided split events at each observed smaller-side size, by source. Sizes not listed have zero events in every source.

Smaller side	Common	Targeted	Structural	Geometric	Seeded
1	0	0	224	1013	204
2	0	0	0	313	64
3	0	4	0	162	18
4	0	0	0	93	9
5	0	0	0	73	6
6	0	0	0	43	4
7	0	4	0	27	1
8	0	0	0	19	1
9	0	0	0	9	0
10	0	0	0	7	0
11	0	0	0	3	0
12	0	0	0	2	0
13	0	0	0	1	0
14	0	0	0	1	0
15	0	0	0	4	0
16	0	0	0	2	0
17	0	0	0	2	0
19	0	4	0	0	0
30	0	0	2	0	0
62	0	0	2	0	0
126	0	0	2	0	0

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